

Technical note

**Electron Spectrometer Using the EUTERPE Magnet
(A.K.A. Dutch Magnet)
for TREX Staging Beam line**

Kei Nakamura

Lawrence Berkeley National Laboratory, USA

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Chapter 1

Introduction

1.1 LOASIS Staging Project

Something about LOASIS staging project comes here, motivate the project.

1.2 Requirements for Electron Spectrometer

Something about requirements comes here.

- **Electron momentum**

Due to the nature of the laser plasma accelerators (LPAs), it is preferred to have as wide momentum acceptance as possible. This specific LPA was designed to produce up to ~ 1 GeV/c of the electron momentum. Therefore, needs to resolve 1 GeV electrons.

- **Momentum resolution**

In order to discuss electron beam quality, it should resolve ~ 1 % level.

- **Angular acceptance**

The wider the better. At least accommodate beams with 1 mrad divergence.

- **Repetition rate**

A nominal repetition rate of 1 Hz operation should be achieved.

- **Charge monitor**

It should serve as the charge diagnostic as well. The calibration has been performed by using a conventional accelerator and verified with the LPA [1].

- **Simultaneous diagnosis of guided laser properties**

The simultaneous observation of guided laser properties must be possible.

- **System dimension**

The system dimension has to be compact enough to fit in the LOASIS experimental A-cave.

1.3 Purpose of this document

The purpose of the document is to provide detailed information of the staging magnetic spectrometer, such as the design philosophy, dimensions of critical hardware, performance estimates and model for the estimates, and the post processing's for analysis.

All information, in terms of files, are located at Dropbox / Server / 4_calibrations + datasheets / manget / stagingDutch.

Chapter 2

EUTERPE Dipole Magnet (A.K.A. Dutch Magnet)

2.1 General information

The magnet was originally designed and fabricated for an electron storage ring EUTERPE [2–5]. The basic parameters are following.

- **Pole dimensions:** 120 mm x 480 mm.
- **Pole gap:** 25 mm.
- **Total weight:** 600 kg.
- **Coil weight:** 20 kg.
- **Coil cross section:** 45 x 80 mm².
- **Number of turns:** 2 x 84 (7 x 12).
- **Conductor:** Hollow copper, 6 x 6 mm² with bore radius of 3.5 mm.
- **Cooling water flow:** 4 liters / minute.

- **Magnetic field:** Maximum magnetic field is 1.5 T for 200 A and 36 V (7.2 kW).
- **Electric properties:** For room temperature, the resistance R is $R \simeq 180 \text{ m}\Omega$.

Photographs of the magnet are shown in Fig. 2.1. We have two identical dipole magnets obtained from the Eindhoven University of Technology (who is the contact?). One was installed in TREX THUNDER beam line and the other was in the TREX Staging beam line. The photos shown in Fig. 2.1 are from the THUNDER beam line.

2.2 Field measurements

The magnetic properties were measured at the magnet test facility in the Advanced Light Source (ALS), on March 2004. A hall probe from Group3 Technology with DTM-141D probe reader was used for the measurement. The hall probe was placed on a linear

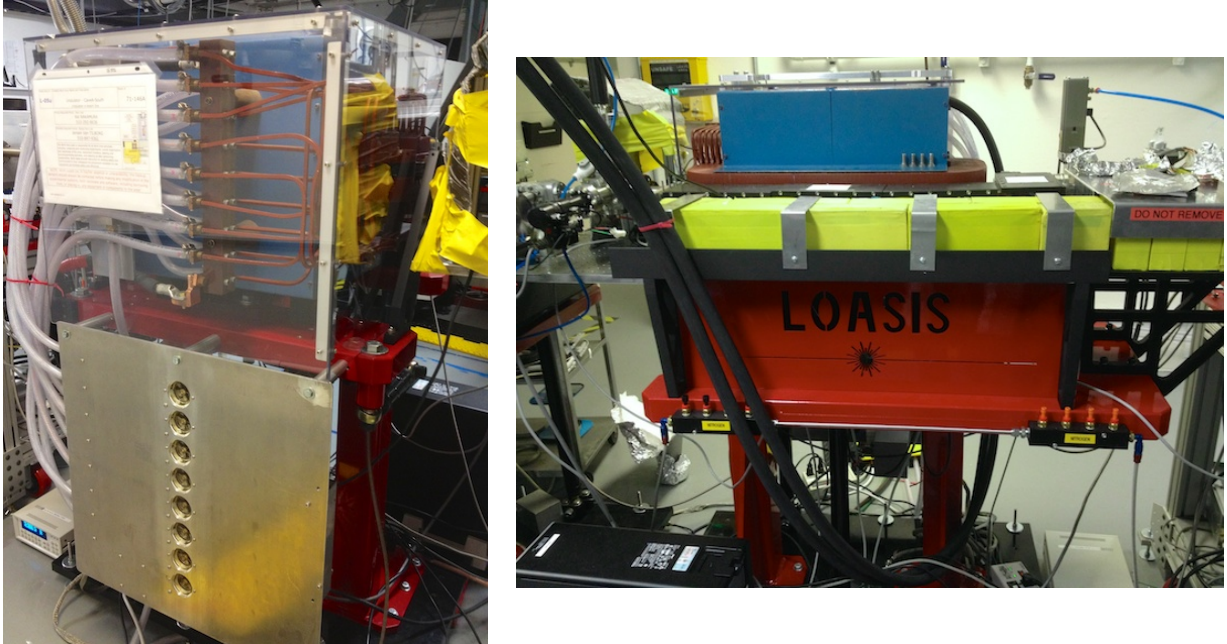


Figure 2.1: Photographs of the Dutch Magnet implemented in the A-cave TREX Thunder beam line as of 2012/11/7.

stage with analog output as the position indicator was used (see Fig. 2.2). All the raw data can be found at `server/4_calibrations+datasheets/magnets/stagingDutch/measurements`. Shown in Fig. 2.3 is the coordinate system of the magnet for this profile measurement. The location $(z, x) = (0, 0)$ was defined as the center of the magnet pole, where z is the longitudinal coordinate with downstream direction being positive, and x is the transverse coordinate. Electrons will be kicked toward positive x direction, where the yoke is open. Note that the magnet utilizes C-shape yoke having an opening for one side. The chamber and scintillating screens is located on opened side.

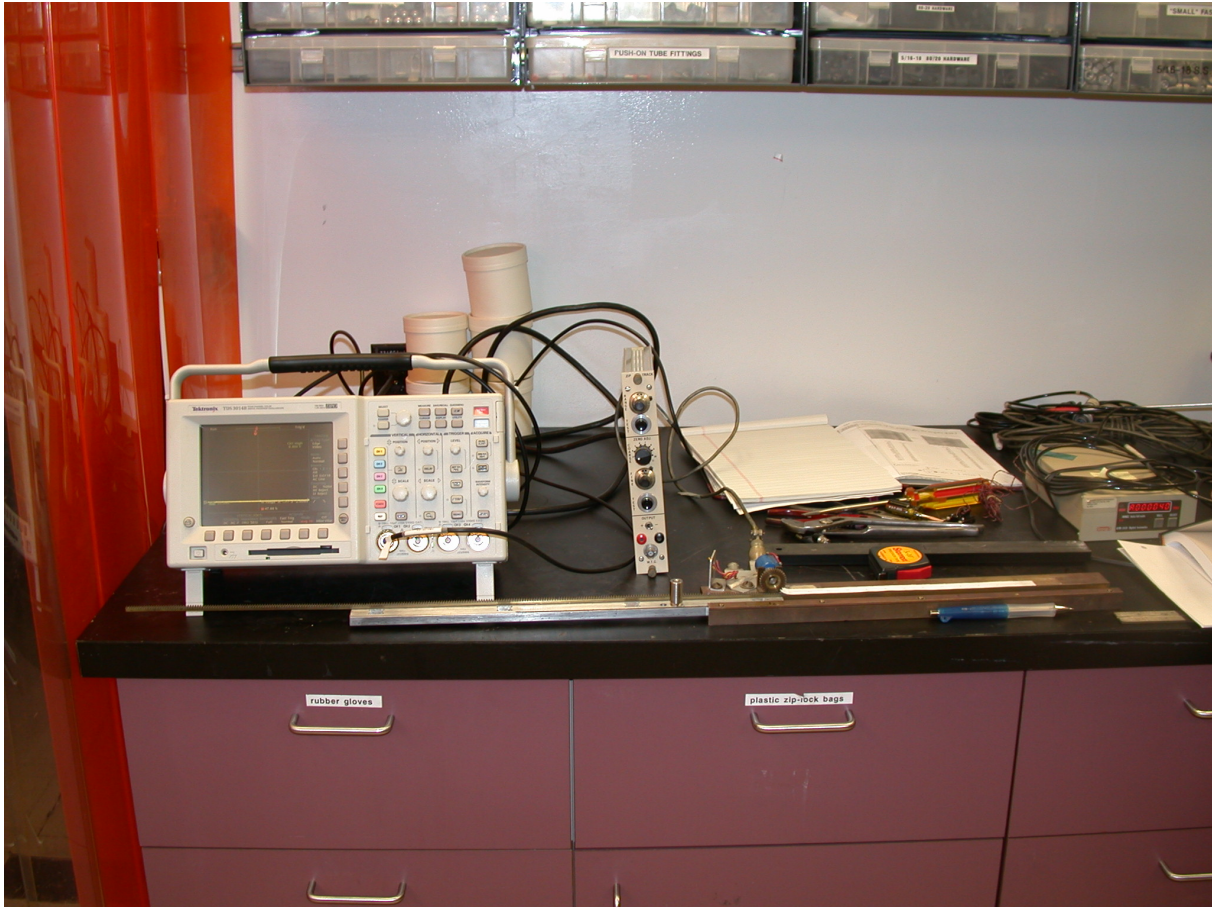


Figure 2.2: Linear stage and hall probe used in the measurements.

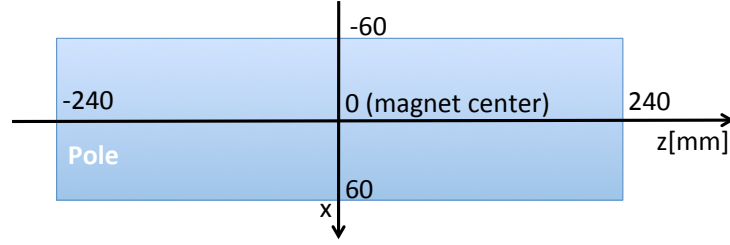


Figure 2.3: Coordinate system for the magnet.

The magnetic field $B(0,0)$ as a function of the applied current I is shown in Fig. 2.4. The magnetic field is linear against the applied current up to 150 A, and the linear fit to the measurement is,

$$B[\text{mT}] = 8.0529 \times I[\text{A}]. \quad (2.1)$$

The maximum magnetic field was about 1509 mT for the applied current of 200 A. The transverse and longitudinal field profile were measured with the excitation current of 50 A, and the results are shown in Fig. 2.5 and 2.6.

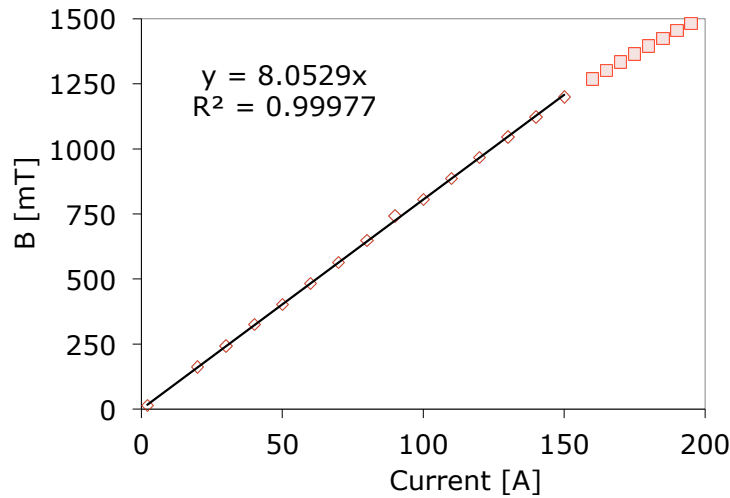


Figure 2.4: Measured magnetic field as a function of the applied current.

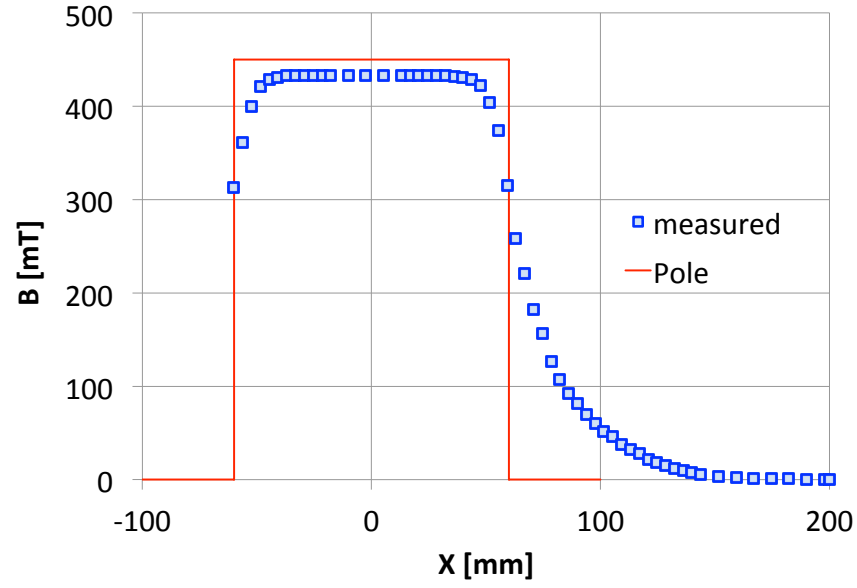


Figure 2.5: The transverse field profile with the excitation current of 50 A (blue squares). Indicated by a solid red line is the physical structure of the magnet pole.

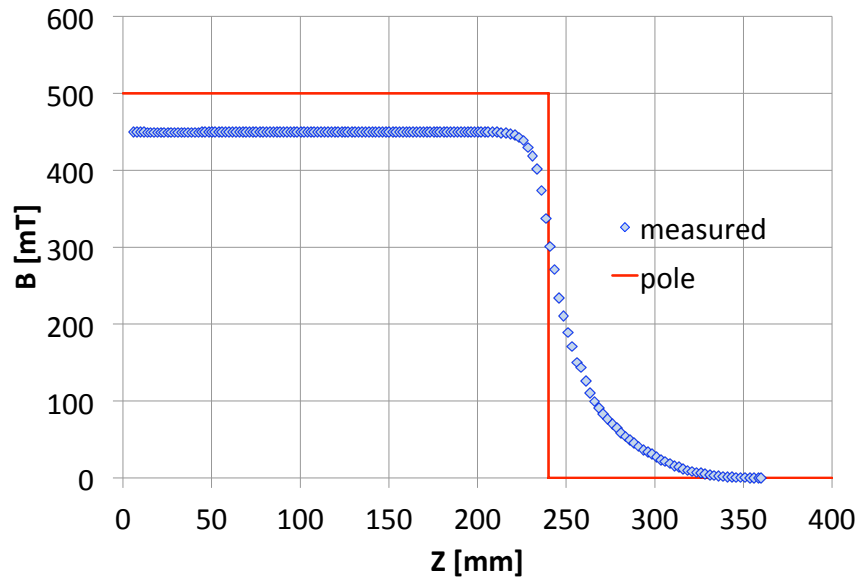


Figure 2.6: The longitudinal field profile with the excitation current of 50 A (blue squares). Indicated by a solid red line is the physical structure of the magnet pole.

Chapter 3

Design and Implementation

Overview of the hardware should be here.

3.1 Main hardware

3.1.1 Support

Talk about the design of the support here.

3.1.2 Chamber

The chamber info should be here.

3.1.3 Observation system

Cameras used for the spectrometer is purchased from Pointgrey [6], FL3-FW-14S3M-C, and the quote is shown in Fig. A.1. Each camera has an identical objective, 8 mm Megapixel Fixed Focal Length Lens from Edmund Optics [7], whose quote is shown in Fig. A.2. Each camera sits on a x-y stage from optosigma [8] Preset Stage 65 x 65 XY, and its quote is attached in Fig. A.3

The first surface mirror used to separate electron beam from the light from the Lanex screen was purchased from Edmund Optics, 4-6 Wave mirror, and the quote is attached in Fig. A.4.

3.1.4 Implementation for the staging beam line

Shown in Fig. 3.1 is a part of the layout for the A-cave staging setup. The magnet is located 2363 mm away from the laser vacuum focus. The choking point for the electron beam transportation is the entrance of the magnetic spectrometer chamber, where the I.D. of the tube is 15.24 mm and 2009 mm away from the laser vacuum focus. The electron beam angular acceptance becomes 1) ± 3.9 mrad for cap1 operation, or 2) ± 4.0 mrad for

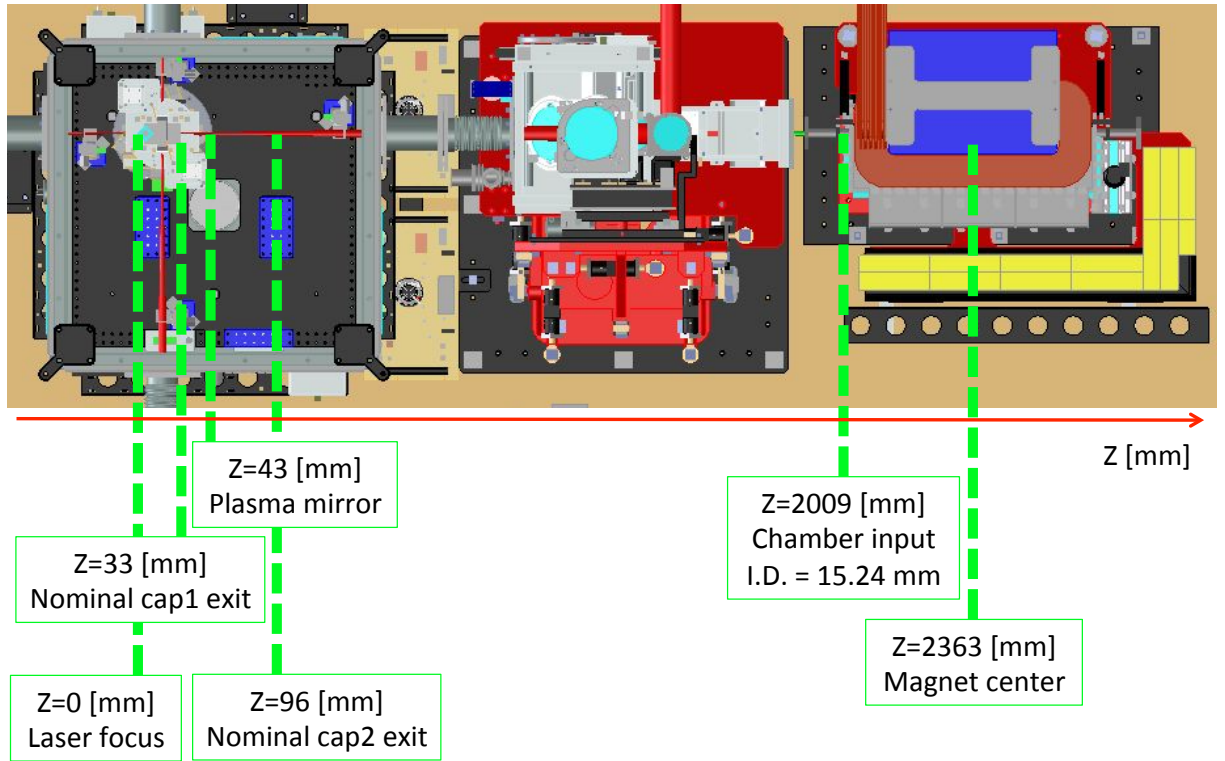


Figure 3.1: A-cave staging setup layout covering from the target chamber to the Dutch magnet.

cap2 operation. Note that those numbers don't included the effect of the magnet to the beam evolution, that is discussed in Chap. 4.

Shown in Fig. 3.2 is the layout of the magnetic spectrometer chamber (screen locations) and the magnet pole location with respect to the chamber (beam line). Note that the magnet is located with a slight offset from the pole center although it was designed to be the center. Here, the coordinate was defined such that $z = 0$ is along the magnet center and $x = 0$ is along the beam propagation axis, instead of the magnet center to be $(z, x) = (0, 0)$. Since the chamber includes the beam line tube, and the screen locations is defined by the chamber structure, the screen locations with respect to the beam line will never change. The magnet location with respect to the chamber is somewhat flexible. Although it was designed such way that the magnet center = the beam line, the magnet location measurement showed the magnet was located 3.3 mm off from the beam line, which is illustrated in Fig. 3.2).

Three CCD cameras were installed to image Lanex screens. The range for each camera was shown in Figs. 3.3 and 3.4. Those images can be found at `server/4_calibrations+`

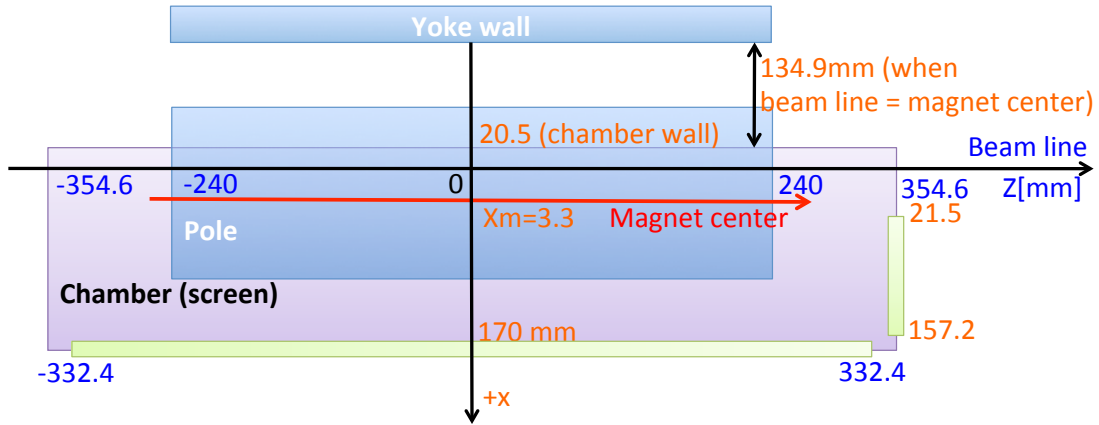


Figure 3.2: Layout of the magnetic spectrometer chamber (screen locations) and the magnet pole location with respect to the chamber (beam line).

datasheets/magnets/stagingDutch/setupSummaries. Due to the wide-angle imaging system to achieve compactness, one has to carefully calibrate the location of the camera. If reference is taken from other plane than where the phosphor surface is, such as screw heads, it would give an error. In this case, the edge of the screen was used for the reference locations based on the technical drawings.

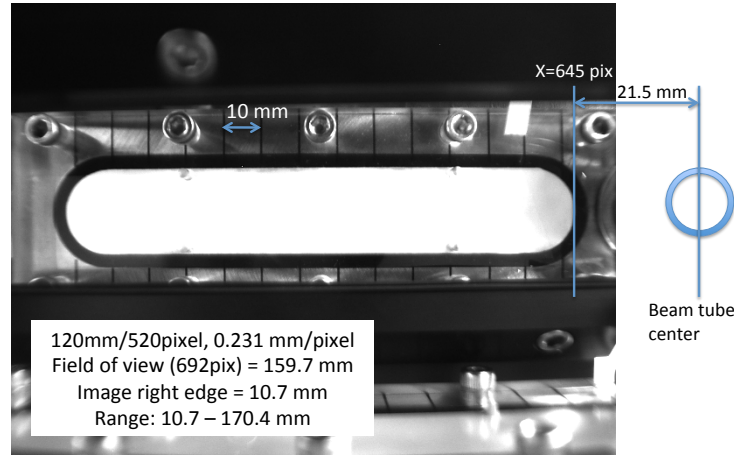


Figure 3.3: Location calibration for the Staging magnetic spectrometer front camera (high energy).

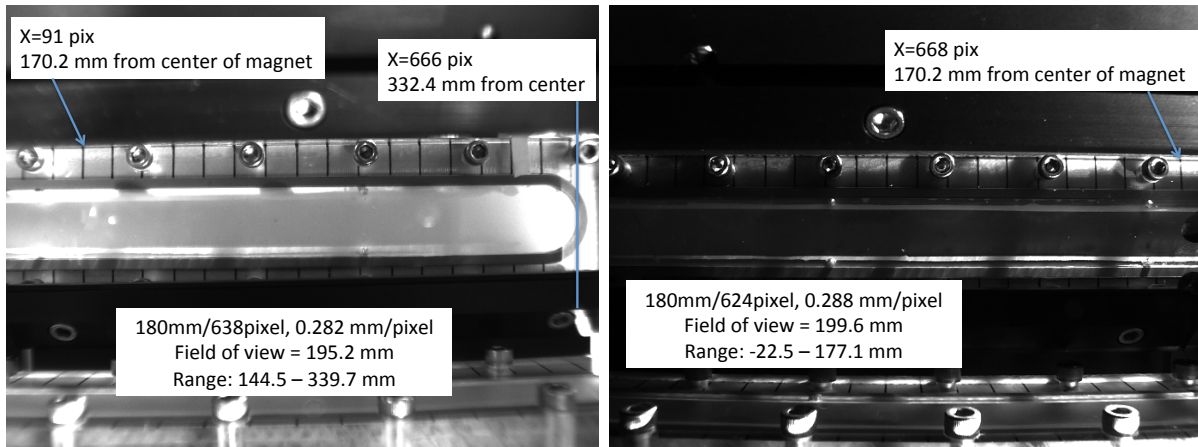


Figure 3.4: Location calibration for the Staging magnetic spectrometer side cameras (top: mid energy, bottom low energy).

3.1.5 Main power supply

Power supply info should be here.

3.1.6 Interlock and control system

Control of the power supply should be talked about some point...

Chapter 4

Spectrometer performance estimate

The performance estimate was carried out in the similar manner with the magnetic spectrometer for TREX system, the varian magnetic spectrometer [9]. The goal is, to tell what this spectrometer can provide, in terms of the momentum acceptance, angular acceptance, momentum resolution, and etc. This Chapter also provides calibration information for a post analysis of the experimental results, where the full momentum spectrum is recovered from the measured images. The steps for a post analysis is described in Chap. 5.

The steps for the performance estimate is following.

1. **Midplane 2D field map**

For the estimate, 2D field map at the mid plane (for dipole, plane includes the beam propagation axis and parallel to the pole surfaces) is required. Based on the measurements shown in Chap. 2, the 2D field map was modeled.

2. **Calculation of reference trajectories**

With the 2D field map, trajectories for each momentum of interest were calculated. For each momentum, this trajectory is the reference trajectory (electron on axis). For each trajectory, required parameters for cosy calculation to obtain beam transfer

matrix, such as the fringe field profile, effective boundary of the field, and etc., were calculated. Also calculated in this step was a part of calibration information for a post analysis.

3. Cosy simulation for transfer matrixes (MAPs)

Based on the trajectory calculation, cosy outputs transfer matrixes in a form of MAP for each momentum. These MAPs are used to calculate beam (many particles) properties at the end of the propagation (screens).

4. MAP analysis

With an assumed the initial particle distribution (beam profile), analysis with MAP gives beam property at the end of propagation (screens). The beam divergence, energy spread and momentum resolution for the spectrometer can be discussed based on this analysis. It also provides a part of calibration information for the post analysis to obtain an electron beam momentum spectrum based on measurements.

4.1 Modeling of the 2D field profile

Based on the field profile measurements described in Sec. 2.2, a 2-D field map was constructed to calculate trajectories. Illustrated in Fig. 4.1 are the measurements (solid red lines), physical pole (blue square) and the modeled area (green square). With a simple multiplication of the transverse and longitudinal profiles, 2-D field map was modeled. It was done by Matlab and the script is located at `server/10_Software/Matlab/Kei/trajectory/dutch/dutchFieldMapV02`. The resultant 2D map is located at `server/4_calibrations+datasheets/magnets/stagingDutch/fieldProfile` in matlab data format, and shown in Fig. 4.2.

Based on the implementation layout shown in Figs. 3.1 and 3.2, a trajectory calculation model was constructed. Shown in Fig. 4.2 by the white solid line are the physical chamber structure, by the red solid lines are the locations of the LANEX screens, and by the black

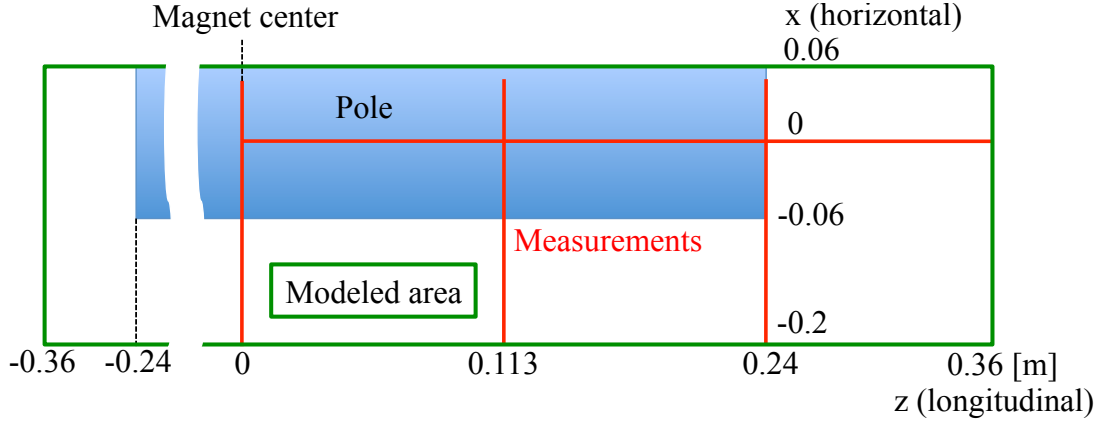


Figure 4.1: Illustration of the modeled area (green square), measurements (solid red lines) and the physical magnet pole structure (blue square).

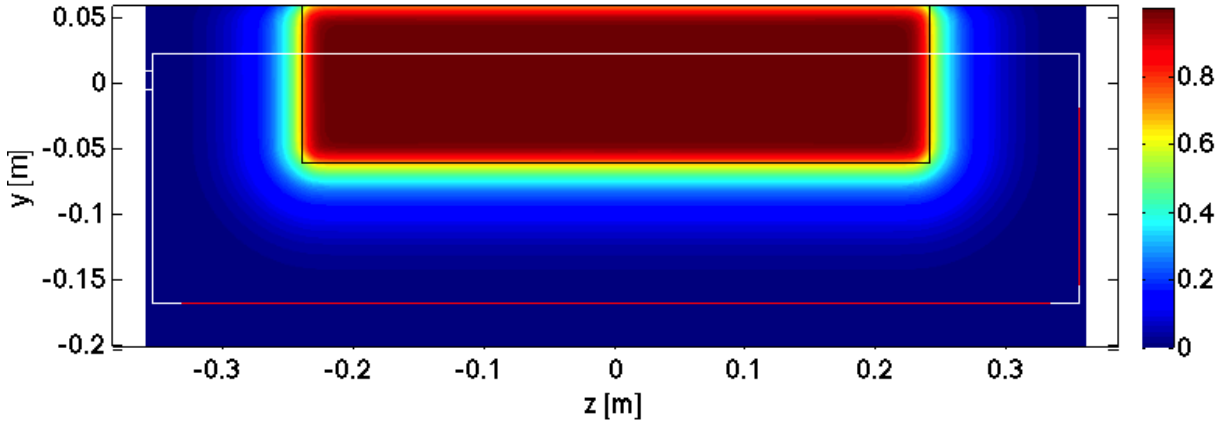


Figure 4.2: Constructed 2-D field map for the Dutch Magnet. The maximum magnetic field was normalized to 1. Also shown by the black solid line is the physical pole structure, by the solid white line is the physical chamber structure, and by the solid red line is the location of the LANEX screens.

solid line is the physical magnet pole structure. In this figure, the longitudinal coordinate 0 is defined as the magnet center, and transverse coordinate 0 is on the center of the beam line, which is 3.3 mm higher than the magnet center as indicated in Fig. 3.2.

4.2 Calculation of reference trajectories

Mid-plane trajectories (trajectories for particles on the propagation axis) were calculated for each momentum using home made matlab script, that is located at `server/Software/Matlab/Kei/trajectory/dutch/dutchTrjV06.m`. The script calculates the trajectories and provides necessary information for the cosy calculation and post analysis as described below.

For cosy calculation, (a) the momentum of the electron for each trajectory has to be given. In order to use an accurate model of the field boundary in stead of the sharp boundary model, (b) fringe fields have to be fitted into the Enge function and the 2 sets of coefficients (for the entrance and the exit of the magnet) have to be provided. Also given to cosy for the fringe field calculation were two extra parameters to accommodate exceptional cases which couldn't be dealt with the default cosy settings. Details is given sections below. In cosy, the beam line was modeled with (c) the drift space 1 (D_1), from the electron beam source to the entrance of the magnet, (d) the magnet, and (e) the drift space 2 (D_2), from the magnet exit to the Lanex screen. For the magnet part, a cosy function DI was used for the estimate. The DI requires following input, (d1) radius of the electron orbit r_e , (d2) bend angle α_e , (d3) magnetic field edge angles for the entrance and exit β_{ent}, β_{ext} , and (d4) the curvature of the field at the entrance and the exit. For this estimate, (d4) was neglected (0 was placed for input) due to the fact that the magnet is rectangular and the effect is small.

For the post analysis of experimental results, following information is required from this trajectory calculation. (α) The electron momentum as a function of the screen location. (β) The total pass length as a function of the screen location. (γ) The incident angle of the electron to the screen as a function of the screen location, that is for the electron beam charge calibration.

4.2.1 Trajectory calculation and momentum (a)

Trajectories were obtained by calculating the equation of motion with the Euler's method. The program can be improved by using the Runge Kutta method in future if necessary. The representative trajectories are illustrated with solid green lines on top of the color scaled magnetic field strength in Fig 4.3. One can see that for 1 T of the magnetic field, the front screen covers from 330 to 2400 MeV/c, and the side screen covers from 11.5 to 270 MeV/c. Note that currently (as of 2013/1/16) the side screen is not fully covered by the cameras (see Sec. 3.1.4 for the camera implementation), covering about 50 MeV/c and above.

All the calculations provided here were done with the peak magnetic field of 1 [T]. Since the beam evolution depends only on the geometrical trajectory, the beam evolution doesn't change as long as one is seeing the same trajectory, but the energy can be changed by the magnetic field. It means that one set of calculations with a specific magnetic field (in this case 1 T) is enough to model this magnetic spectrometer. Although different strengths of the magnetic field would be used depending on the electron beams in experiments, one can obtain necessary information for analysis by simply multiplying the field strength of

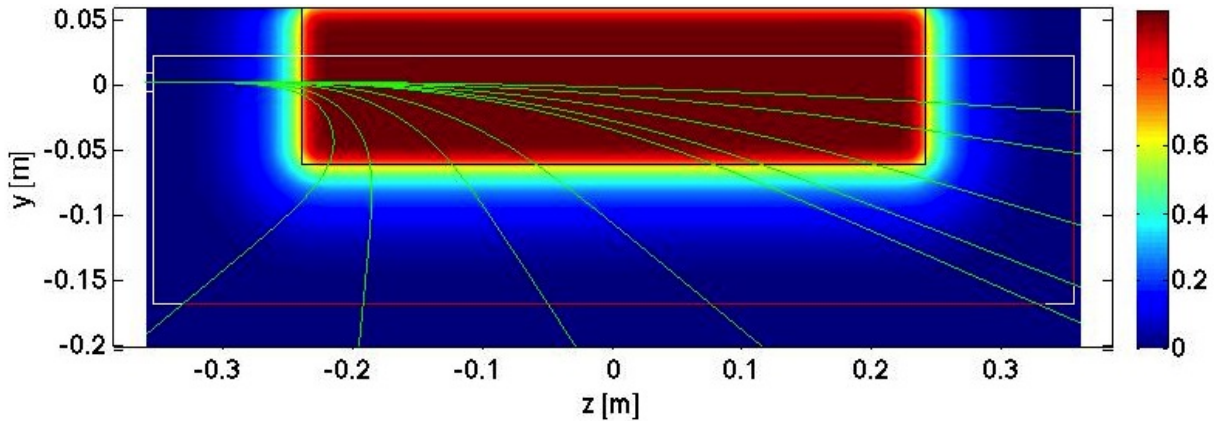


Figure 4.3: Representative trajectories. From higher energy, 2400, 1000, 500, 330, 270, 100, 50, 20 and 11.5 MeV/c electrons.

the experiment to the momentum of this estimates.

For each trajectory, the cosy calculation requires the momentum of electron p_e [MeV/c] so that the energy dependent beam evolution can be discussed along the analysis of one referenced trajectory. For this evaluation, it is not necessary to specify the momentum because the trajectories are calculated for each energy. Nonetheless, the electron momentum was given with the magnetic field normalized to the max field B_{pk} being 1 [T]. Note that the momentum is linear to the magnetic field, while the total energy E_e and kinetic energy E_{eK} are not as can be seen from following equations,

$$E_e[\text{eV}] = \frac{\gamma m_e c^2}{e}, \quad (4.1)$$

$$E_{eK}[\text{eV}] = E_e - E_{e0}, \quad (4.2)$$

where the electron rest mass energy $E_{e0} = m_e c^2 / e \simeq 0.511$ [MeV/c] and m_e is the rest mass of an electron.

$$p^2 = \gamma^2 m_e^2 v^2 = (\gamma^2 - 1) m_e^2 c^2, \quad (4.3)$$

$$p_e[\text{eV}/c] = \frac{pc}{e} = \sqrt{\frac{\gamma^2 m_e^2 c^4}{e^2} - \frac{m_e^2 c^4}{e^2}} = \sqrt{E_e^2 - E_{e0}^2}, \quad (4.4)$$

$$= \sqrt{E_{eK} (E_{eK} + 2E_{e0})}.$$

The Lorentz force is linear to the magnetic field and velocity of the charged particle, and the electron orbit radius r_e is given as follows.

$$r_e[\text{m}] = \frac{p}{eB[\text{T}]} = \frac{p_e[\text{eV}]}{cB}. \quad (4.5)$$

When $E_e \gg E_{e0}$, it becomes $E_e \sim E_{eK} \sim p_e$. For a discussion of highly relativistic electron beams, differences among the total energy, the kinetic energy and the momentum are negligible. One has to be careful when it is not highly relativistic. Since the magnetic spectrometer can be operated for relatively low energy beams, discussions here are focused

on the "momentum" taking advantage of the its linearity to the magnetic field strength.

4.2.2 Foci estimates

By calculating two trajectories with the same electron momentum but with different angles, one can simulate a beam envelope in bending plane, that gives you rough estimate of the focus location for each momentum. Shown in Fig. 4.4 are the beam envelope calculation for 30, 84, 200 and 800 MeV/c beams. The beam was assumed to have 0 size at the exit of the cap1 ($z=33$ mm, see Fig. 3.1), and divergence was assumed to be ± 3 mrad. One can see that the beam focuses before it reaches screen for 30 MeV/c, roughly focuses at screen for 84 MeV/c, focuses after the screen for 200 MeV/c, and is almost collimated for 800 MeV/c beams.

One can obtain the best momentum resolution by placing screen to follow foci, which was done for the varian spectrometer design [9]. The staging spectrometer chamber design was a complete copy of the Thunder undulator beam deflector / spectrometer (Undulator Dutch), in order for the cost performance. That spectrometer was optimized not for the best momentum resolution performance but optimized the best beam angular acceptance,

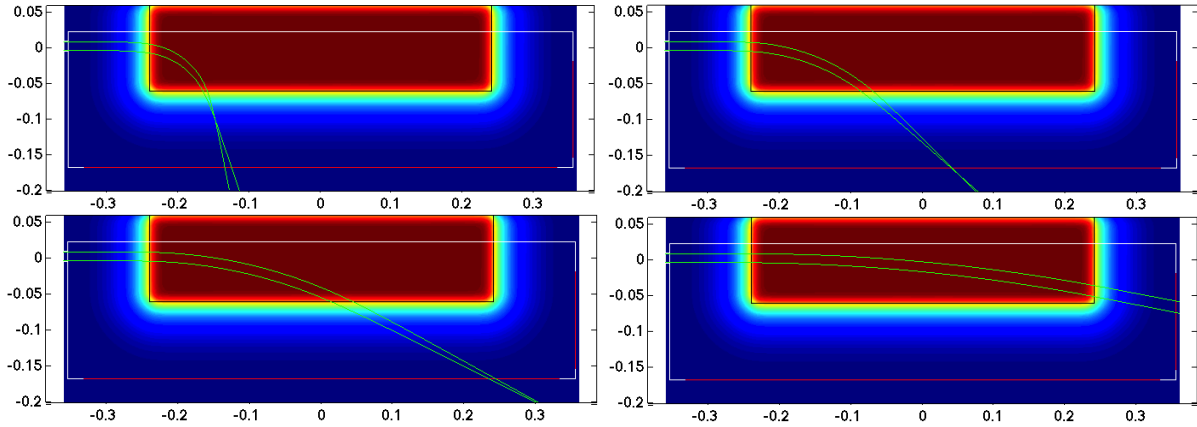


Figure 4.4: Simulation of the beam envelope for 30 (top left), 84 (top right), 200 (bottom left) and 800 MeV/c (bottom right) with the magnetic field of 1 T.

because it was located about 7 m away from the beam source. This is the reason that the screen location is not following the foci of the electron beams.

4.2.3 Fringe fields (b)

Shown in Fig. 4.5 left is an example of a trajectory calculation and analysis. The solid blue line shows the field profile for the 1000 MeV/c electron for $B_{pk} = 1$ [T]. Indicated by the dashed black lines are the upstream effective boundary, center of the magnetic field profile, the downstream effective boundary, and the location of the boundary $+ 5 \times \text{gap}$, where the effective field boundary is described in the next Section. Shown by the solid red and green lines are where the screen is located and where the field becomes 0 [T], respectively.

For a trajectory analysis to have a set of cosy input parameters, at first, the center of the field profile S_{mc} is obtained to separate upstream and downstream profile because the

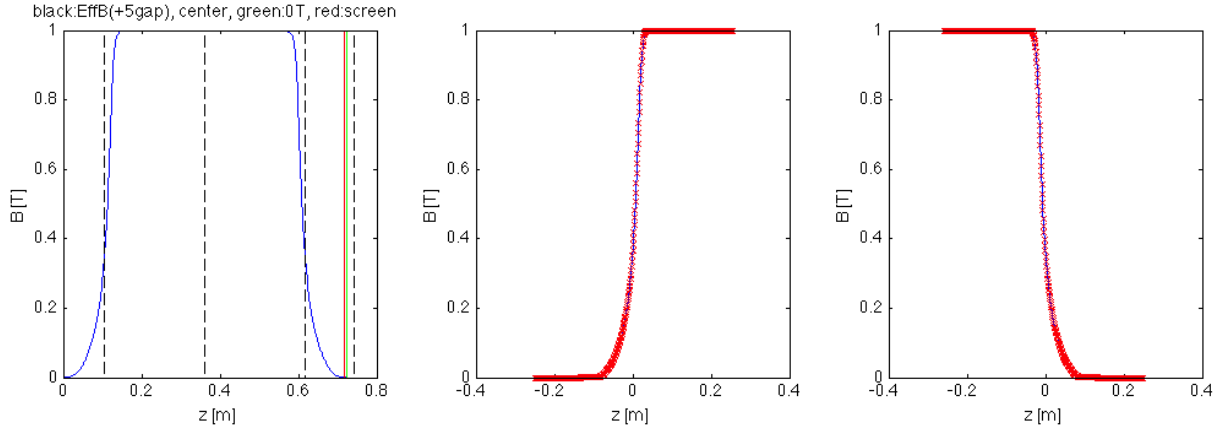


Figure 4.5: Left: the field profile for the 1000 MeV/c electron (solid blue). Indicated by the dashed black lines are the upstream effective boundary, center of the magnetic field profile S_{mc} , the downstream effective boundary, and the location of the boundary $+ 5 \times \text{gap}$. Shown by the solid red and green lines are where the screen is located and where the field becomes 0 [T], respectively. Shown in center and right are the results of the fitting into the Enge function.

each fringe field need to be fitted into the Enge function separately. Shown in Fig. 4.5 center and right are example of the fitting for the upstream and downstream fringe fields, respectively. Although the cosy can calculate MAPs without the fringe field given by the Enge function by using sharp boundary mode, the most accurate mode of cosy calculation is the one uses this fitted fringe field description. For this performance estimate, the calculated field profile was fitted into the Enge function,

$$B(s) = \frac{1}{1 + \exp[a_1 + a_2(s/g) + \dots + a_6(s/g)^5]}, \quad (4.6)$$

where a are the Enge coefficients, s is the coordinate of electron propagation, and g is the gap of the dipole magnet which is 25 mm for the Dutch magnet. By using the FR function of the cory, one can specify the fringe field for the input and output of the dipole magnet.

When the fringe filed mode is enabled in cosy calculation by using the FR function, it calculates in default the range from $-3 \times g$ to $5 \times g$, where g is the gap of the poles and negative direction is toward inside of the magnet. This default setting can become problem for following two cases, (1) fringe extends above $5 \times g$, and (2) screen is located

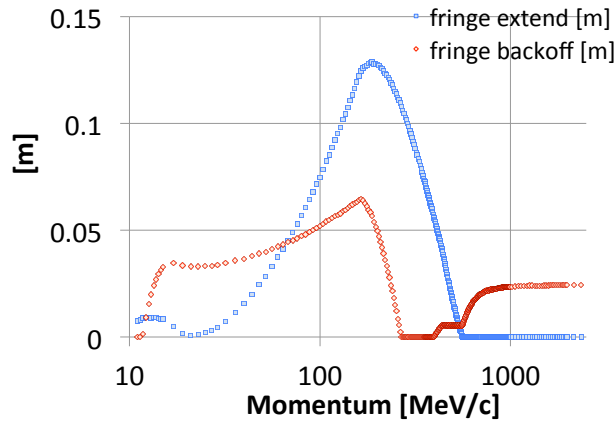


Figure 4.6: Fringe extend and fringe back-off as a function of the electron momentum.

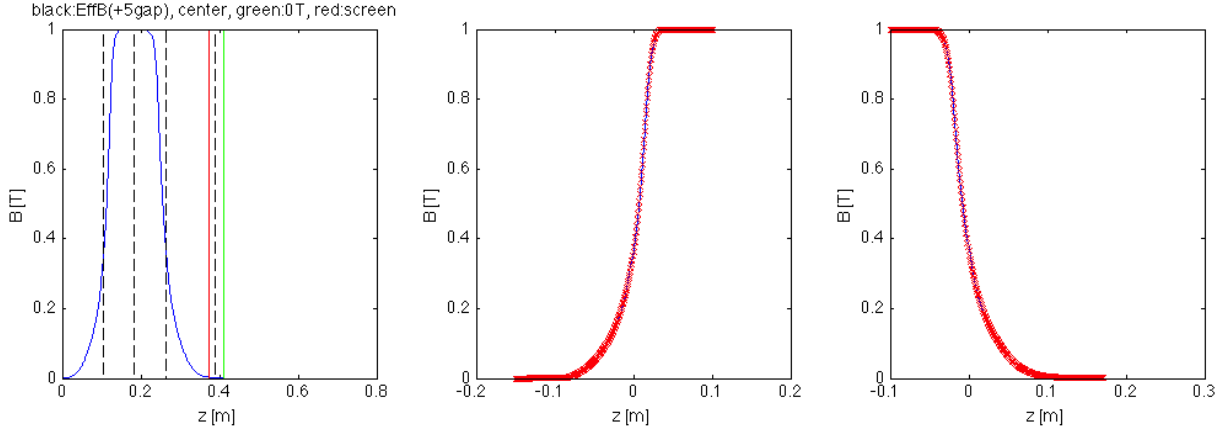


Figure 4.7: Calculated field profile and fitting results for 45 MeV/c electron. See Fig. 4.5 for explanation of the lines. In this example, one can see that the magnetic field ends at further than effective boundary + $5 \times$ gap, and the screen is inside of the fringe field.

inside of the fringe field. For the case of (1), the fringe calculation has to be extended to cover down to the location where $B = 0$. For this, a parameter fringe extend F_{ext} was calculated and shown in Fig. 4.6. For the case of (2), the calculation has to be shortened so it stops the calculation at where the screen is located. The parameter for this, fringe back-off F_{bo} was calculated and shown in Fig. 4.6. One can see that the fringe calculations had to be extended for the electron momentum below about 500 MeV/c. But for lower energies, from 12 to 65 MeV/c, the fringe back-off exceeds the fringe extend, meaning the calculation was shortened because the screen was located inside of the fringe field. Such example, 45 MeV/c case, is shown in Fig. 4.7. One can see that the end of the field (solid green) is well beyond the $5 \times g$ line (dashed black), but the screen (solid red) is located before the $5 \times g$ line.

4.2.4 Drift spaces (c, e)

For each trajectory, the peak field B_{pk} was obtained, and used to evaluate the effective boundary and the equivalent electron orbit radius (see next Section). Note that B_{pk} for the whole filed profile being 1 [T] does not necessary mean that the B_{pk} for each trajectory

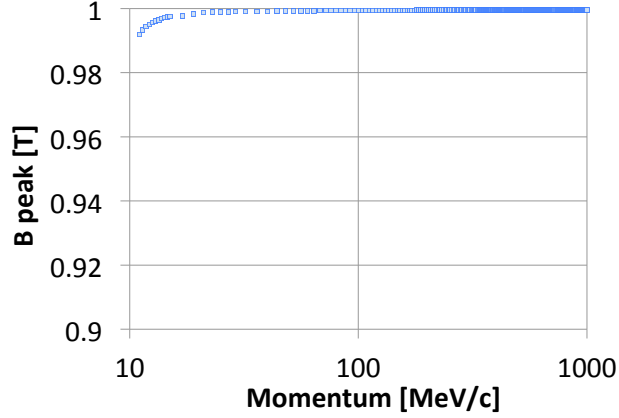


Figure 4.8: The peak field B_{pk} as a function of the momentum.

is also 1 [T]. For trajectories of relatively low energy electrons, it can be deflected before it reaches the $B = 1$ region. The peak field as a function of the momentum is shown in Fig. 4.8.

By integrating the field along s , electron propagation coordinate, and normalizing with B_{pk} , one can obtain the effective length of the magnet L_{eff} ,

$$L_{eff} = \frac{\int B(s)ds}{B_{pk}}, \quad (4.7)$$

and shown in Fig. 4.9 left. As expected from Fig. 4.3, the longest effective length is seen about 800 MeV/c trajectory. From the effective lengths for the upstream and downstream, and the location of the center S_{mc} , one can obtain the location of the effective magnet boundaries, that are indicated by the dashed black lines in Fig. 4.5 left. The drift space 1 for the cosy input D_1 is given by the distance from the electron beam source to the upstream effective boundary, and it was almost constant, 2076 mm for cap1 configuration. The drift space 2 is obtained by the distance from the downstream effective boundary to the screen, and shown in Fig. 4.9 right. One can see that there is a distinct feature at around 290 MeV/c where the front (high energy) screen and the side (low energy) screen

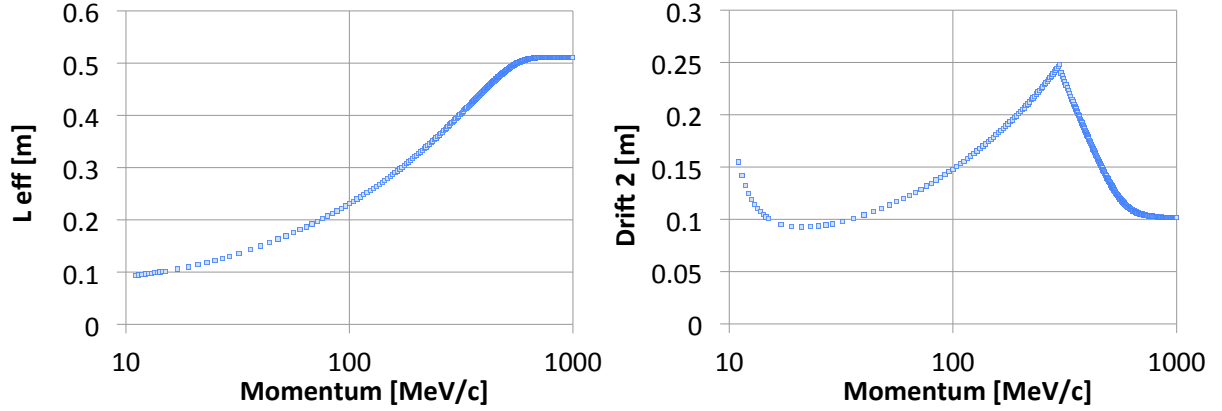


Figure 4.9: The effective length, L_{eff} , in left and the drift 2, D_2 , in right as a function of the momentum, respectively.

are separated. For the front screen, lower energy trajectories have longer drift 2 because it has to travel to the farther corner of the chamber. For the side screen, lower energy trajectories have shorter drift 2 as the distance between the magnet and screen gets closer.

4.2.5 Magnet parameters (d)

Although the radius of the electron orbit r_e is a function of the field strength, one can get a constant r_e , equivalent electron orbit radius to the case of the sharp boundary model, and that is what cosy calculation needs,

$$r_e[\text{m}] = \frac{p_e[\text{eV}/c]}{c[\text{m/s}]B_{pk}[\text{T}]} \quad (4.8)$$

With the effective length of the magnet for each trajectory, then one can calculate the bend angle of the electron α_e for each trajectory,

$$\alpha_e[\text{rad}] = \frac{L_{eff}[\text{m}]}{r_e[\text{m}]} \quad (4.9)$$

Both of the parameters as a function of the electron momentum are shown in Fig. 4.10.

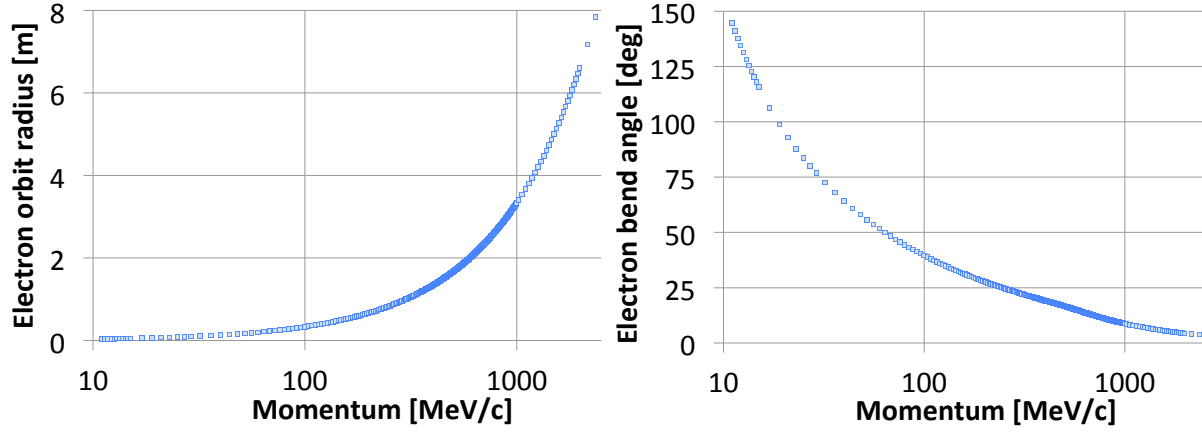


Figure 4.10: The equivalent electron orbit radius, r_e , in left and the electron bend angle, α_e , in right as a function of the momentum, respectively.

The electron incident angle to the magnetic field at the entrance and exit affects "edge focusing effect". In cosy, it was defined such way that the positive angle de-focuses beams in bending plane, and 0 degree is the normal incident angle and no edge focusing. The incident angles were calculated from the angle between the electron and 2D field gradient of the magnetic field at the effective boundaries, and shown in Fig. 4.11. For the input

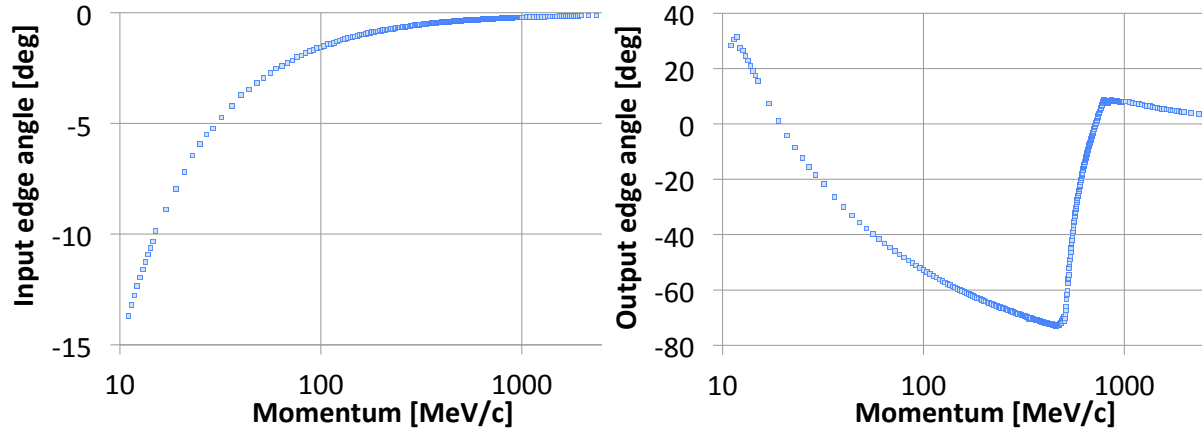


Figure 4.11: Input (left) and output (right) edge angle of the electrons as a function of the momentum, respectively.

incident angle, one can see that it is almost 0 degree for high energy electrons. It is because high energy electrons go almost straight to the magnet, while lower energy particles get bent by the fringe fields resulting in having an angle against the magnetic field gradient at the effective boundary. For the exit incident angle, very high energy particles have almost 0 angle with the same reason as the input incident angle. It drastically changes where the orbit hits the apex of the magnet, and then eventually becomes 0 degree when the electron exits from the side of the magnet with normal angle.

4.2.6 Calibration information for a post analysis (α, β, γ)

For a post analysis of experimental results, the electron momentum and the incident angle of the electron to the screen as a function of the screen location were calculated, and shown in Fig. 4.12. The front screen coordinate is defined such way that the 0 is the laser beam axis, and the side screen coordinate is defined such way that the magnet center is 0. Note that the momentum here is for the case of $B_{pk} = 1$, meaning that one has to multiply the momentum with the measured magnetic field for the post analysis to obtain

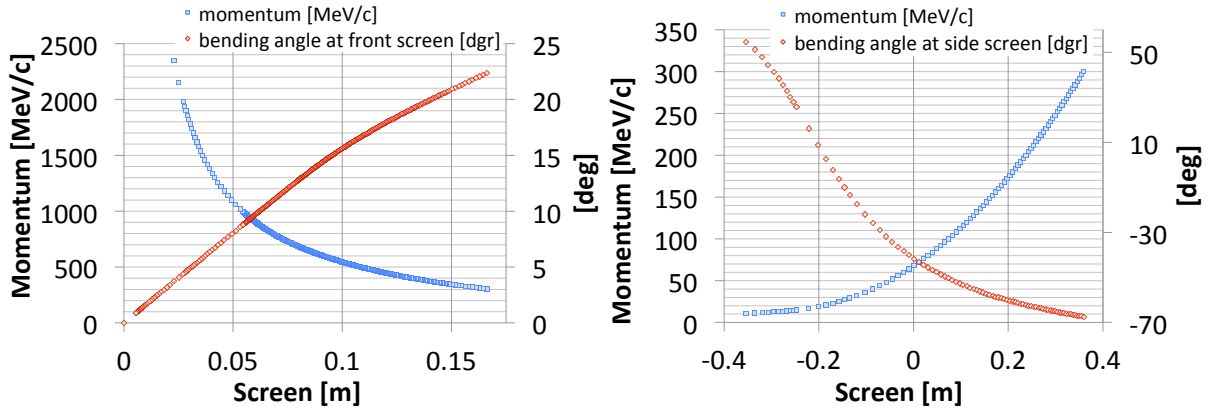


Figure 4.12: Electron momentum and the incident angle of the electron to the screen as a function of the screen location. Left: front screen, Right: side screen. The front screen coordinate is defined such way that the 0 is the laser beam axis, and the side screen coordinate is defined such way that the magnet center is 0.

the momentum. The bending angle at the screen defines the electron - screen interaction length. When the charge of the beam is estimated, one has to take into account this length to normalize. Shown in Fig. 4.13 is the total path length, from the source to the screen, as a function of the screen coordinates. The beam acceptance and divergence can be estimated by the total path length and MAP analysis shown in the Sec. 4.4. One can see that the total path length is from about 2.3 to 2.7 m for the spectrometer. Considering the opening of the chamber being 15.24 mm, the angular acceptance is about ± 3 mrad neglecting the diverging/converging force from the magnet.

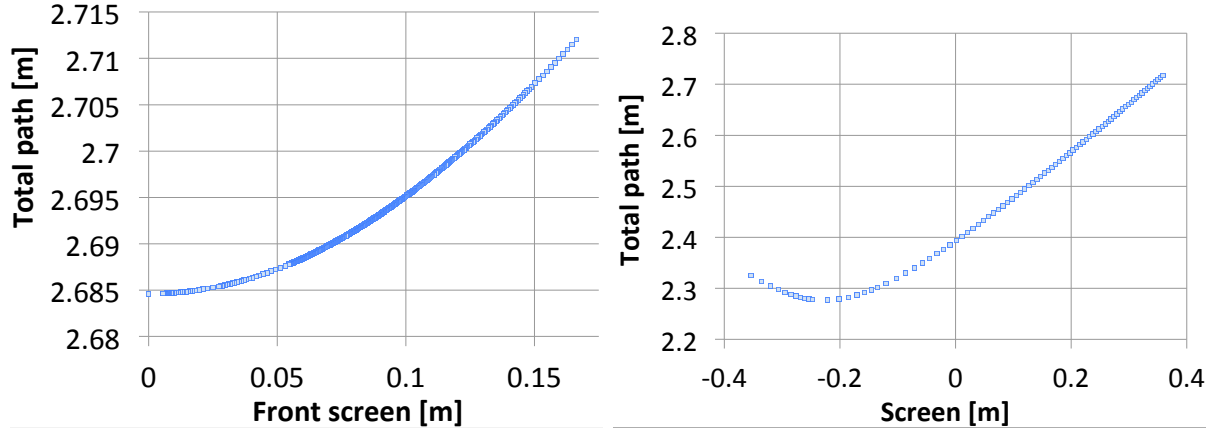


Figure 4.13: Total path length as a function of the screen coordinates, left for the front screen coordinate, and the right for the side screen coordinate.

4.3 MAP calculations with cosy infinity

In order to estimate the beam evolution along the propagation, the beam transfer optics code cosy infinity (cosy) [10] was used. The cosy outputs the result as a form of MAP. Shown in Fig. 4.14 is an example of the MAP for the first order case. The MAP consists of 6 columns and many rows. The first column is for x , then goes x' , y , y' and l at the 5th column. The numbers in 6th column indicate the order for each coefficients, from left x , x' , y , y' , l , and dp/p . Here, x is the coordinate for bending plane, and y is the coordinate

-2.822351	-12.50205	0.000000	0.000000	-0.3868419	100000
-5.830411	-26.18104	0.000000	0.000000	-0.8249190	010000
0.000000	0.000000	16.57908	83.33586	0.000000	001000
0.000000	0.000000	33.91846	170.5538	0.000000	000100
0.000000	0.000000	0.000000	0.000000	1.000000	000010
0.7276345E-01	0.1852535	0.000000	0.000000	-0.3951450E-01	000001

Figure 4.14: The MAP example (first order).

for non-bend plane. The s location of a particle is indicated by l , and the momentum spread is given by dp/p . For example, if you see the first column, see three non-zero terms. For each non-zero term, you consult to 6th column to see what is contributing to. Then one can construct an equation to describe x as

$$x_1 = -2.822351x_0 - 5.830411x'_0 + 0.07276345dp_0, \quad (4.10)$$

where notation 0 indicates input parameters while notation 1 indicates output. An input consists of a set of parameters $[x_0, x'_0, y_0, y'_0, l_0, dp_0]$ and so does the output, except dp never changes without accelerating (or decelerating) components, which cosy doesn't deal with.

For this estimate, the beam line was modeled simply with three blocks, drift 1, the dipole magnet and drift 2. As discussed in the previous section, all input parameters were calculated by the matlab script. With cosy, the MAP was calculated for each trajectory with up to third order, and the script and the MAPs were located at `server/4_calibrations+datasheets/magnets/stagingDutch/cosy`.

4.4 MAP analysis

A matlab script was made to read MAPs provided by the cosy calculation, and simulate beam evolution. The script is located at `server/Software/Matlab/Kei/trajectory/mapAnalysis/dutchMapAnalysisV02.m`. The script generates a beam with an arbitrary

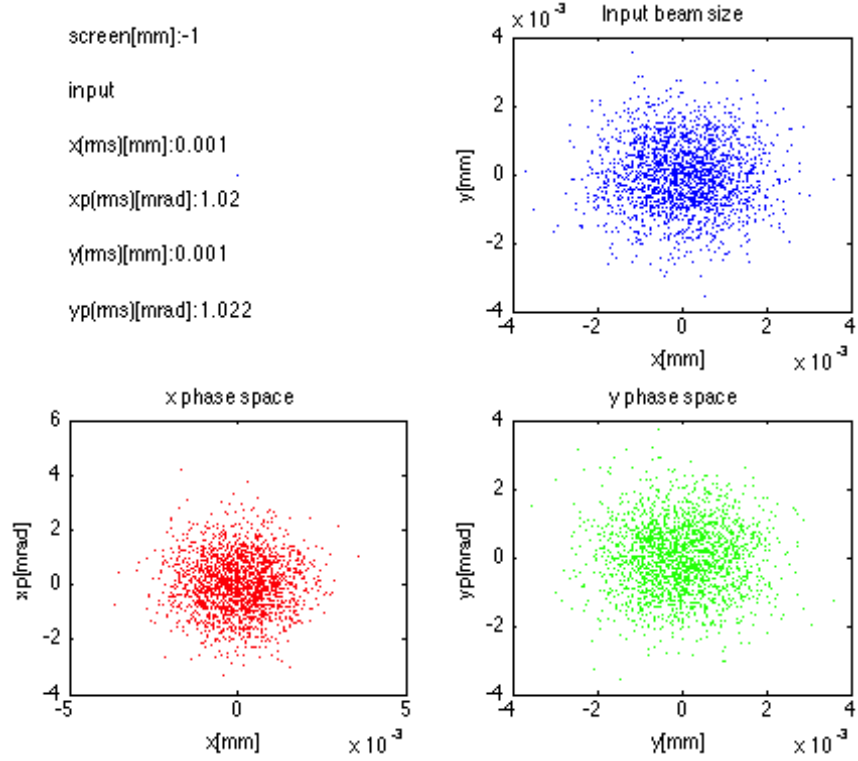


Figure 4.15: Example of the input beam. At the beam source, the beam has $\sigma_{x0} = \sigma_{y0} = 1$ [μm], and $\sigma_{x'0} = \sigma_{y'0} = 1$ [mrad].

number of particles randomly distributed with Gaussian profile in x , x' , y , and y' . An example of the generated beam is shown in Fig. 4.15, where the input parameters are $\sigma_{x0} = \sigma_{y0} = 1$ [μm], and $\sigma_{x'0} = \sigma_{y'0} = 1$ [mrad].

With the MAPs for each trajectory and input beam parameters, the beam distribution at the screen for each trajectory (each mono energetic beam) can be calculated. The results for the representative trajectories, 30, 84, 200 and 800 MeV/c beams are shown in Fig. 4.16. One can see that quite strong effects of the focusing in x (bending) and defocusing in y (non-bending) planes for low energy beams, while the propagation becomes almost ballistic for high energy beams such as 800 MeV/c. For these calculations, the beam size at source was assumed to be 1 μm , and the beam divergence 1 mrad, and the

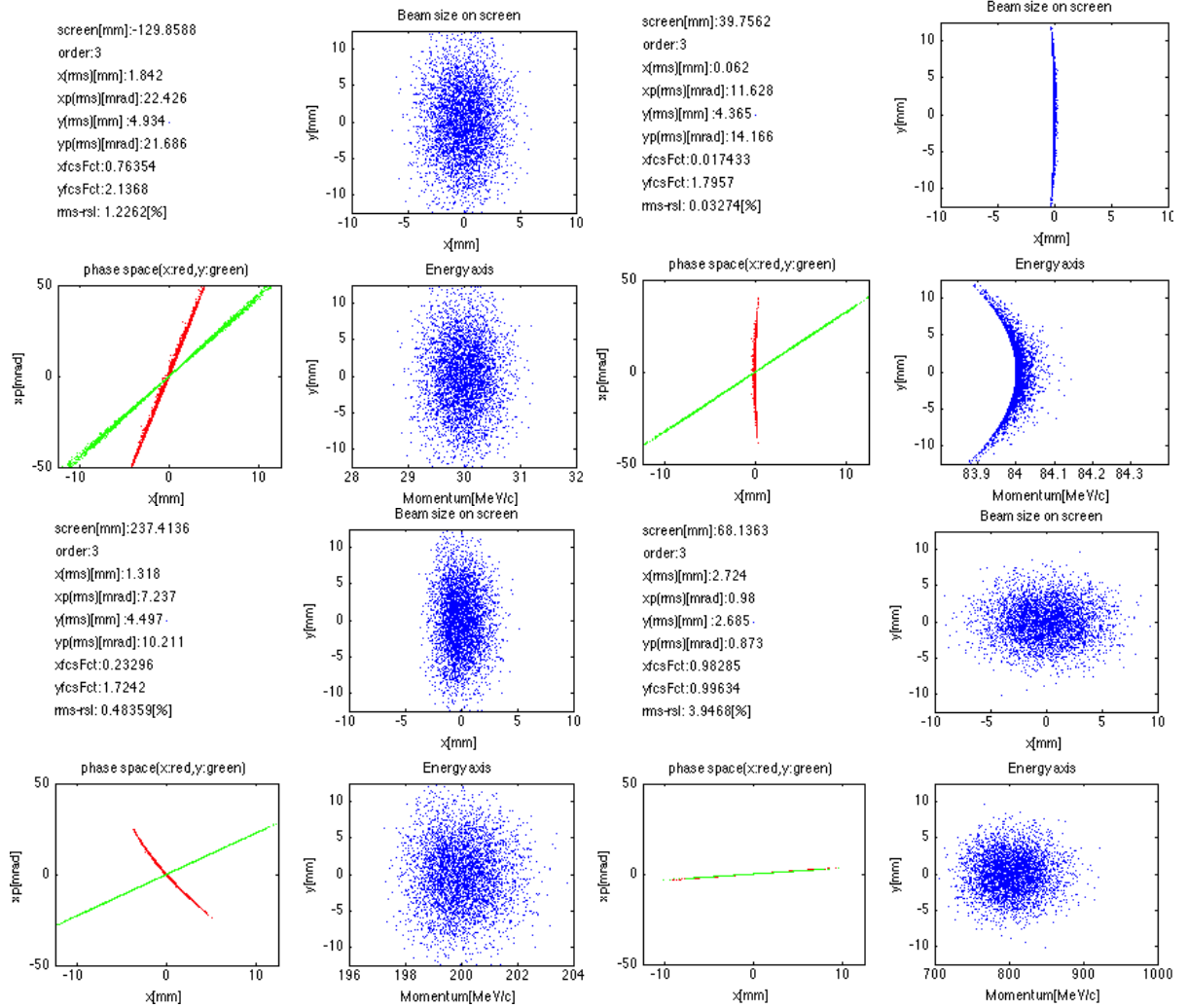


Figure 4.16: The beam distribution at the screen for 30 MeV/c (top left), 84 MeV/c (top right), 200 MeV/c (bottom left) and 800 MeV/c (bottom right). The input beam distributions are $\sigma_{x0} = \sigma_{y0} = 1 \text{ } [\mu\text{m}]$, and $\sigma_{x'0} = \sigma_{y'0} = 1 \text{ [mrad]}$.

beam source z-location was assumed to be at the exit of cap1 ($z = 33 \text{ mm}$, see Fig. 3.1). Note that the focus trend seen here agrees well with the rough focus estimated done by the trajectory calculation (see Fig. 4.4).

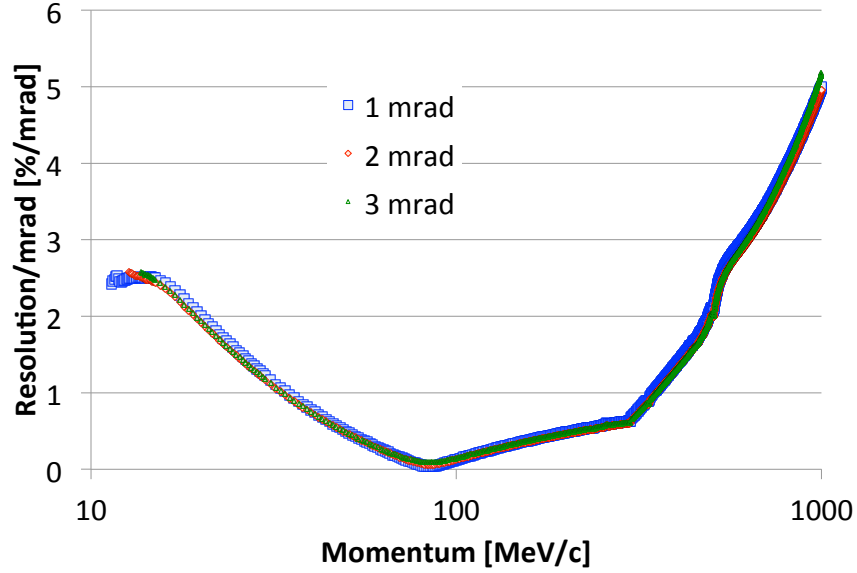


Figure 4.17: The normalized momentum resolution as a function of the momentum. Input parameters are $\sigma_{x0} = \sigma_{y0} = 1$ [μm], and $\sigma_{x'0} = \sigma_{y'0} = 1, 2$, and 3 [mrad].

The momentum resolution for each momentum can be estimated by calculating apparent energy spread of the mono-energetic electron beam. Since there is no slit for this magnetic spectrometer, the momentum resolution depends on the beam divergence. In any of the fig. 4.16, on the bottom of the top left quadrant, there is the apparent momentum spread of the mono-energetic beam noted as "rms-rsl". The spectrometer can not resolve momentum spread below this number. The momentum resolution, or the apparent momentum spread of the mono-energetic beam for each momentum, is shown in Fig. 4.17 for the beam divergence of 1, 2, and 3 mrad in σ (standard deviation). The momentum resolution was normalized by the input beam divergence. One can see that the resolution is below 1 % for 1 mrad divergence beam from 35 to 340 MeV/c, which accords to the entire side screen currently covered with the cameras.

To the first order, the momentum resolution should be linear to the input divergence. Since the normalized momentum resolution is almost on top of each other from 1 to 3

mrads divergence beams, one can say that the beam evolution is almost linear for this parameter regime. It simplifies analysis for the momentum resolution. In order to obtain the momentum resolution, one doesn't have to simulate all the input beam divergence. Simply by multiplying the beam divergence with the energy resolution given for 1 mrad input divergence case, one can estimate the momentum resolution.

As has been shown above, the beam generally gets converged in the bending plane by the magnet, giving the focusing and enhancing the momentum resolution. On the other hand, the beam is more diverged by the magnet, narrowing the angular acceptance. The angular acceptance in non-bending plane can be estimated by this MAP analysis. First, the opening at the screen is 15.4 mm. The second, total path length for each momentum was calculated in Sec. 4.2 as shown in Fig. 4.13. The total length gives the angular acceptance for the ballistic case, and that was about ± 3 mrad. By taking the beam evolution effect into the analysis, one can estimate the beam angular acceptance as shown in Fig. 4.18. For higher momentum electrons, such as above 800 MeV/c, the path length dominantly decides the beam angular acceptance. One can see that for lower momentum electrons, the defocusing force severely limits the angular acceptance. The acceptance is about 1/4 of the Varian magnetic spectrometer [9]. Note that the acceptance in bending plane can be wider, because the chamber has wider space horizontally to disperse in momentum. It was estimated to be ± 3.9 mrad based on the I.D. of the magnet input tube as discussed in the Sec. 3.1.4.

All the information discussed in the Sections from 4.2 to 4.4 is compiled into text files "StagingMagCap1.txt" and "StagingMagCap2.txt", and they can be found at `server/4_calibrations+datasheets/magnets/stagingDutch/ESMCalib`. The file ends with "Cap2" includes calibration information for the case of cap1 run, while the other provides the case for cap2. Post processing requires these calibration text files, so the copy of them are located at `stagingFileServer/data/ESMCalib` for convenience.

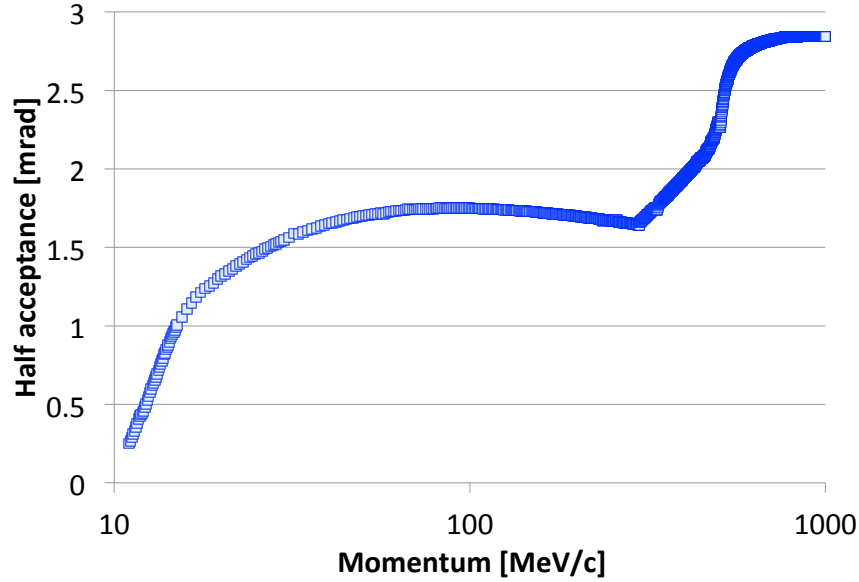


Figure 4.18: The angular acceptance of the non-bend plane as a function of the momentum. For the estimate, the input electron beam parameters are $\sigma_{x0} = \sigma_{y0} = 1$ [μm], and $\sigma_{x'0} = \sigma_{y'0} = 1$ [mrad].

4.5 Performance Estimate Summary

The summary of the performance estimate for the staging magnetic spectrometer using the Dutch magnet is presented below.

- **Momentum acceptance**

The spectrometer has a potential to cover from 11.5 to 2400 MeV/c, a factor of 200, in a single shot as shown in Fig. 4.3. Note that this is the case of the peak magnetic field 1 [T]. For the current setup (as of May 8, 2013), which utilizes three cameras to cover a part of the screens as shown in Figs. 3.3 and 3.4, it covers from 60 to 2400 MeV/c, a factor of 40.

- **Momentum resolution**

The momentum resolution depends on the beam divergence. For 1 mrad beam divergence, high resolution measurements (less than 1 % momentum resolution)

can be achieved from 35 to 345 MeV/c as shown in Fig. 4.17. In terms of the camera views of the current setup (as of May 8, 2013), it covers whole 2 side camera views.

- **Angular acceptance**

The angular acceptance in the bending (horizontal) plane is estimated to be ± 4 mrad as discussed in Sec. 3.1.4. For non-bending (vertical) plane, it depends on the momentum as shown in Fig. 4.18. For the high resolution region, it is roughly ± 1.7 mrad, and for the high energy region it is about ± 2.8 mrad. Note that compared to the varian spectrometer that has the angular acceptance of about ± 12 mrad, the acceptance of the staging spectrometer is only 1/8. Compared to a round pole magnet, a rectangle magnet introduces sharp wedge focusing effect at the exit boundary, resulting in the better momentum resolution while losing the angular acceptance.

- **Error estimate**

The estimate for the error in determining the absolute momentum due to the pointing fluctuation has not been performed yet. To be done.

Chapter 5

Post Processing and Analysis

During experiments and for post experimental analyses, one needs to convert raw images to somewhat useful format so that can optimize accelerators, discuss electron beam properties, and then eventually create viewgraphs for publications. In order for those purposes, several Matlab scripts were written. In this chapter, those scripts are briefly discussed to help users so that they can conduct quick analysis, optimize accelerator performance, and/or conduct in-depth analyses by themselves. All the scripts discussed here can be found at `server/10_Software/Matlab/Kei/TREX/Staging`.

1. Post Processing Script

A post processing script "stgDutchV02.m" is to provide (1) calibrated image: whose count is directly calibrated to the charge [aC], all three images are combined, the horizontal axis is converted to momentum with the uniform momentum spread dp , and the vertical axis is converted to angle with the uniform angle spread $d\theta$. (2) text file with horizontal axis information, and (3) text file with vertical axis information. It also generates a text file "ScanXXX_MagSpecPostP.txt" for each scan, with charge information for each camera. With those information, in principle, one can perform any post analysis without actually knowing about the magnetic spectrometer calibrations. So, this script has to be executed after experiments unless the equivalent post process is provided by other means. Those three outputs are

located at `date/analysis/scanXXX/StagingMagnet`, in the folders named "intImg (integrate image)" for the image, "engTxt (energy axis text file)" for the horizontal axis information, and "trvTxt (transverse axis text file)" for vertical axis information.

2. Simple batch analysis with viewgraph

A script "stgDutchViewV01.m" reads files created by the post processing script (not raw images), and provide (1) a viewgraph in a format that one could put in a presentation, and (2) text file "ScanXXX.MSAnalysis.txt" with simple analysis result such as peak / mean momentum and angle, angular divergence, momentum

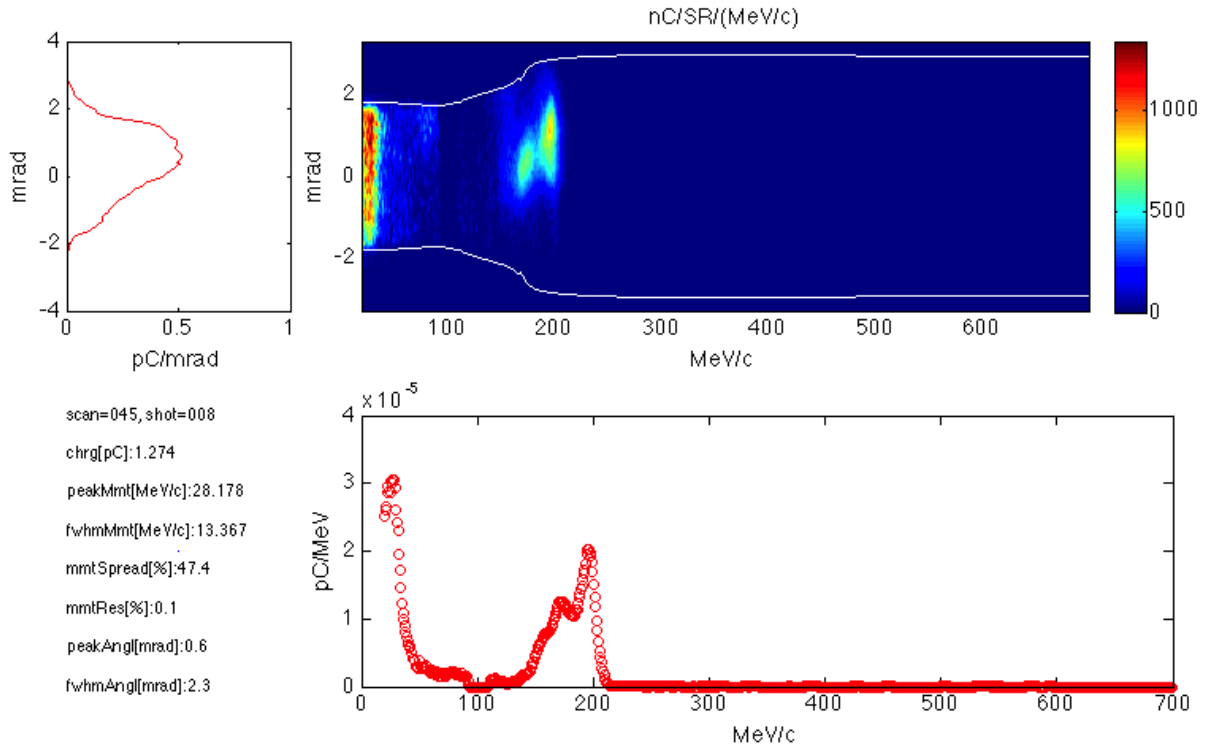


Figure 5.1: Example of the viewgraph that generated by the script "stgDutchViewV01.m". It consists of the transverse profile (momentum integrated) at the top left, 2D charge density profile at the top right, the momentum spectrum (angle integrated), and the results of a quick analysis at the bottom left.

spread, and momentum resolution. It processes for whole one scan or multiple scans. The output is located at `date/analysis/scanXXX/StagingMagnet`, in the folders named "viewImg (viewgraph image), and example is shown in Fig. 5.1.

3. Simple single-shot analysis with viewgraph

A script "stgDutchViewV01.m" is a single-shot version of the script described above. It can specify ROI for the momentum axis. It can be useful when one want show zoomed-in images in a presentation. The output is located at the identical place with the identical name as the batch analysis script, meaning the viewgraph will be overwritten. An example is shown in Fig. 5.2.

4. Simple Statistical Analysis with Viewgraph

A script "stgDutchSttAnlV01.m" carries out a simple statistical analysis, by read-

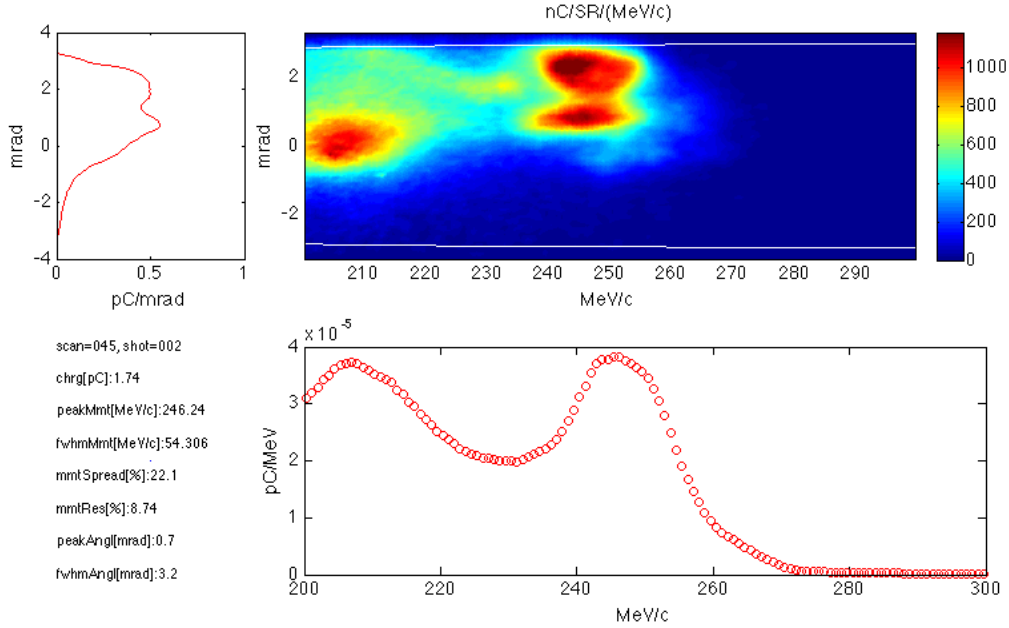


Figure 5.2: Example of the viewgraph that generated by the script "stgDutchSingle-ViewV01.m". The format is identical to Fig. 5.1. One can see that the momentum axis is zoomed into 200 - 300 MeV/c range.

ing processed images and axes produced by the post processing program and log file created during the experiment. It calculates mean and standard deviation for given bin of the scanned, or not scanned (for example input energy with natural fluctuation) parameter. An example is shown in Fig. 5.3. The mean transverse

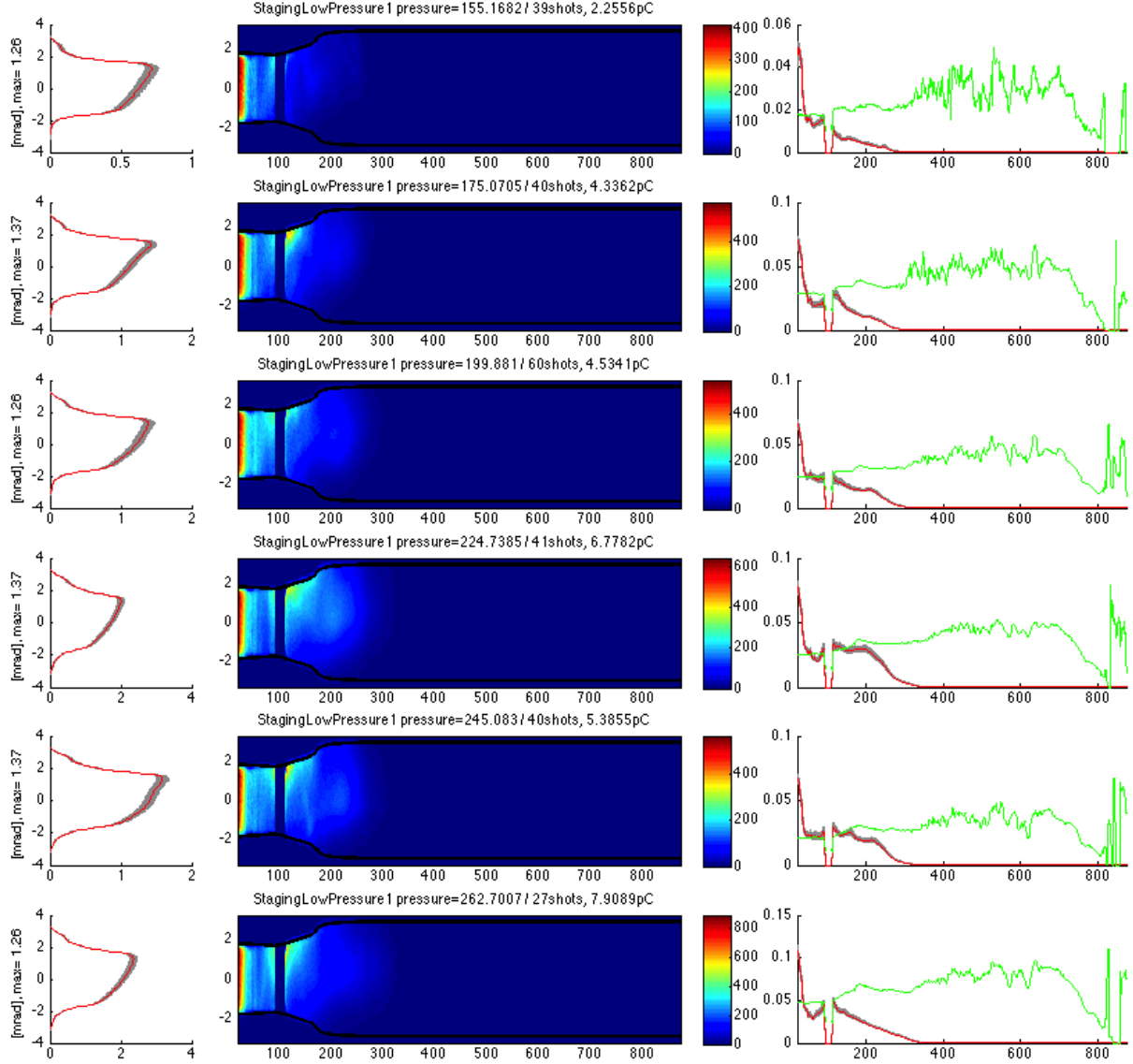


Figure 5.3: Example of the simple statistical analysis. Details are described in the manuscript.

beam profile is shown in the left by solid red curve with its standard deviation by greyed area. An averaged 2D angular-momentum charge density is shown in the middle, with scanned parameter and average charge in its title. Shown in the right side is the angle - integrated momentum spectrum by solid red line with its standard deviation by greyed area.

5.1 Post Processing Script

This section describes what the post processing script does briefly.

5.1.1 Output files

First, it outputs three kinds of file for each shot, and one summary file for each scan. The details of outputs are following.

1. Charge calibrated 16bit Integer image

It is 16 bit png image, with its count is directly calibrated to the charge in the unit of aC. The number of the pixels is 128 in vertical (angular) axis and 1024 in horizontal (momentum), where the intervals for both the momentum axis and angular axis are uniform. Note that with the uniform interval, it is straightforward to normalize to the charge to charge density, such as aC/MeV/c and aC/mrad.

Since it is just an image, it does not contain the axis information other than the pixel number. In order to retrieve horizontal axis information, one has to read the text file with the horizontal axis information, and the same goes for the vertical axis.

Although one can obtain the angular beam profile and the momentum spectrum by integrating over the horizontal axis and the vertical axis, respectively, those information are already saved in the text files. This integer image is useful when one needs to do any image processing and / or ROI for in-depth post analysis.

2. Horizontal axis information text file

The text file under "engTxt" folder contains the horizontal axis information. It consists of a title header in the first row and 1028 (row) x 7 (column) numerical information. The first column is the momentum in MeV/c, then the second the charge in fC, the charge density in fC/MeV/c, acceptance angle in mead, the electron incident angle to screen in degree, the normalized momentum in MeV/c/T, and the momentum resolution in %/mrad. The first column is also the horizontal axis information for the integer image described above.

When one needs to plot a spectrum, one should show the first column as horizontal axis, and the third column in the vertical axis (not the second column). The fourth column is to show the half - angle acceptance of the system. Since the spectrometer diverges the electron beam depending on the trajectory and the travel length for each momentum is different, the angular acceptance has the momentum dependence. The electron incident angle is necessary when one needs to estimate the solid angle. The sixth column is the momentum when the magnetic field is 1 T. This information allow users to estimate the magnetic filed, and other detailed information for in-depth analysis. The momentum resolution depends on the momentum and the input divergence. Since the last column is normalized by the input divergence in mrad, one has to multiple this value by the divergence for the momentum of your interest.

3. Vertical axis information text file

The text file under "trvTxt" folder contains the vertical axis information. It consists of a title header in the first row and 128 (row) x 3 (column) numerical information. The first column is the angle in mrad, then the second the charge in fC, and the charge density in fC/mrad. The first column is also the vertical axis information for the integer image described above.

4. Post processing summary text file

The summary text file is located at the `analysis/ScanXXX/StagingMagnet`, and it consists of a title header in the first row and the number of shots in the scan (row) x 6 (column) numerical information. The first column is the scan number, then the second the shot number, the logic for saved experimental image, the charge from camera 1, camera 2, and camera 3 in pC. The logic in the third column shows non zero number when an image was not found to be saved. This can be used for filtering the bad shot out for post analysis.

5.1.2 Input files

Second, in order to process raw images into a processed image and other outputs shown in the Sec. 5.1.1, it requires information, that are provided via text files. Following is the list of the input files and brief description.

1. Three raw images

Experiments save three images from the magnetic spectrometer cameras, the location of which are at `StagingFileServer/data/day/scans/ScanXXX/MagCam1`, `MagCam2`, `MagCam3`. The `MagCam1` is the camera for the front screen (high momentum). They are 16 bit png image (significant bit depth is 12). The camera configuration is shown in Sec. 3.1.4. The post processing script requires background images. They are saved at the same folder with negative shot number.

2. Scan data file

For each scan, a scan data file is saved at `StagingFileServer/data/day/scans/ScanXXX/ScanDataScanXXX.txt`, that contains the information for the magnetic field for each shot. The title of the column is 'StagingMagSpec Bfield'. The post processing script reads this log file and extract the magnetic field information.

3. Camera calibration text file

A text file contains the camera calibration information is located at `StagingFileServer/data/ESMCalib/yy_mmddcamCalib.txt`. For location calibration, it has "FOV [mm]"

(field of view) and "Left edge [mm]". The coordinate is shown in Fig. 4.2. The cameras can have ROI for saving. In order to compensate vignette, this ROI information has to be passed on, and they are saved in this file as "ROI Y offset", "ROI height", "ROI X offset", and "ROI width". The terminology is identical to the one from MAX, National Instruments. To align images each other in height, the vertical center pixel has to be specified for the post processing, that is saved as "Y center" in this file. One can further ROI for the post processing, that are specified as "Y Start", "Y End", "X Start", and "X End" in this file. Lastly, one can rotate raw image to align well, that is specified as "rot [deg]" in this file. When one change either 1) location of any camera, 2) saving ROI, or 3) post process ROI, this file needs to be updated. The post processing script finds the calibration dated closest to the day of experiments processing among the files dated earlier than the day of experiment. Shown in Fig. 5.4 is `130417camCalib.txt`, that was used to process 2013 04 18 experiment.

cam number	FOV [mm]	Left edge	ROI Y offset	ROI height	ROI X offset	ROI width	Y center pixel	Y Start	Y End	X Start	X End	rot [deg]
1	159.7	170.4	162	130	0	692	59	21	97	55	650	-0.2
2	195.2	144.5	180	130	0	692	69	35	100	1	680	-0.8
3	199.6	-22.5	184	150	0	692	84	50	115	1	579	0.2

Figure 5.4: `130417camCalib.txt`

4. Lanex calibration text file

This file contains Lanex calibration information, which is used to obtain the calibration factor to convert camera count to the charge. See gdoc "TREXClb Lanex calibration" for details. The staging Dutch magspec is specified as setting number 6. The parameters "FOV slope" and "FOV offset" provides conversion information to estimate the distance from the screen to the camera, which is critical for the charge calibration. For each setting (there are 7 as of 2013/5/7), the vignette effect was experimentally measured and the parameters to compensate are listed as "vignette 4", "vignette 2", and "vignette 0". Also measured for each setup experimentally is

the distance dependence for the sensitivity. The parameters "sensitivity 2", "sensitivity 1", and "sensitivity 0" provides the curve for sensitivity. Since camera can be ROI for saving, the post processing script needs to know the original number of pixel for the camera, that are shown as "full width" and "full height" in this file. Shown in Fig. 5.5 is `1303111anexCalib.txt`, that was used to process 2013 04 18 experiment.

setting number	FOV slope	FOV offset	vignette 4	vignette 2	vignette 0	sensitivity 2	sensitivity 1	sensitivity 0	full width	full height
1	0.7315	22.753	2.17E-12	-1.76E-06	1	1.81E-05	0.00116156	0.01691097	512	384
2	0.9121	24.762	3.83E-12	-3.64E-06	1	2.47E-05	1.46E-03	1.74E-02	512	384
3	0.1896	0.982	2.85E-12	-1.41E-06	1	3.38E-07	-9.36E-05	0.03011273	512	512
4	0.5721	7.7876	2.55E-12	-9.29E-07	1	1.57E-05	-0.0007053	0.05496356	512	384
5	0.592	6.9601	5.42E-14	-4.24E-07	1	1.83E-06	-5.49E-05	-5.43E-05	1288	964
6	0.7866	10.824	5.66E-12	-3.12E-06	1	5.43E-06	-0.0002585	0.0051	692	516
7	0.9865	27.171	1.55E-13	-6.31E-07	1	4.73E-06	5.55E-05	0.00477781	1384	1032

Figure 5.5: `1303111anexCalib.txt`

5. Spectrometer calibration text file

This file, for example `StagingMagCap1.txt`, contains the calibration information for this magnetic spectrometer, that was described in Chap. 4. The file consists of 54 columns, and the post processing scripts read following 6 columns. (1) "momentum [MeV/c]" contains the momentum. (2) "side screen" or "front screen" contains the position at the screen, whose coordinate is shown in Fig. 3.2. By combining (1) and (2) with the magnetic field information, one can do position-momentum conversion. (3) "bending angle at side screen" or "bending angle at front screen" indicates the incident angle of the electron to the screen. It defines the interaction length of the electron with the screen, therefor critical to the charge calibration. (4) "total path" is the total path length (from the source to the screen) of the electron. It is needed to estimate the beam divergence and acceptance. (5) "y conversion fct" shows the beam is diverging/converging with respect to the ballistic trajectory (drift in vacuum). It is needed to estimate the beam divergence and acceptance. (6) "momentum rsl [%/mrad]" shows the normalized momentum resolution. Although

this script doesn't do any analysis, it passes this information onto the output text file (horizontal axis information file) so that one can obtain the momentum resolution at post analysis.

5.1.3 Process

Here describes process the script does, with all input information described in Sec. 5.1.2.

1. **Back ground subtraction**

The first process is background subtraction. Should any damage pixel exist in the cameras, it should be taken care of it here.

2. **Low pass filter**

When high energy photon hits the sensor, it usually creates high count in very small area. In order to remove those very localized high counts pixel, the low pass filter is applied.

3. **Vignette compensation**

Vignette, less collection efficiency at the corners of the image, is compensated.

4. **Image rotation**

Due to the imperfect setup, camera can be slightly rotated with respect to the system. It is fixed here.

5. **Incident angle effect compensation**

This is a part of the charge calibration. The incident angle of the electron to the screen defines its interaction length with the screen. The light yield is normalized to the normal incident angle yield.

6. **Count to charge conversion**

For the charge calibration, all spacial and location dependence are compensated by process above. Here, finally one can convert count to charge.

7. Binning in horizontal axis

For three images with 100 x 700 resolution, simple concatenation yields a image with 100 x 2100 resolution. Since the output image resolution is 128 x 1028, there is sizable compression in horizontal axis. In addition, more compression occurs in cameras for lower energy side because the output image has the uniform momentum step. One can see from Fig. 5.1 that the first camera (high momentum) covers 80 % of the output image (above 100 MeV/c), and other two cover only 20 %. For this compression, one can get more noise with a simple interpolation. Therefore, non-linear binning was performed before an interpolation to convert image with uniform momentum step. It is also done before an interpolation for the vertical axis to save the time for calculation.

8. Vertical axis conversion and interpolation from position to angle

In order to stitch images together, they have to have common vertical axis. For convenience for post analysis, here the vertical axis (location) is converted to the angle with uniform spacing, $d\theta$. It can be done by the 2D interpolation. Here, image was reshaped to long 1D array and 1D interpolation was done to make process faster.

9. Combine image

With the common vertical axis, images can be stitched each other into one big image.

10. Horizontal axis interpolation for uniform momentum step

Lastly, in order to output the image with uniform momentum step, interpolation was performed for horizontal axis. As in the vertical axis interpolation, it was done through 1D interpolation to save calculation time.

5.2 Other Scripts

Other three scripts are quite simple. It basically reads three output files from the post processing script, and plot them. Note that one doesn't have to read any calibration files such as Lanex calibration, Camera calibration, and Spectrometer calibration file for post analysis.

Appendix A

Quotes



Flea3

Ultra-Compact IEEE 1394b Cameras



Part Number	Description	QTY 1 Price	QTY 10 Price
FL3-FW-03S1M-C	SONY ICX618 1/4" CCD; mono; 648 x 488; 120 fps; C-mount	\$565 USD	\$525 USD
FL3-FW-03S1C-C	SONY ICX618 1/4" CCD; color; 648 x 488; 120 fps; C-mount	\$565 USD	\$525 USD
FL3-FW-03S3M-C	SONY ICX414 1/2" CCD; mono; 648 x 488; 76 fps; C-mount	\$845 USD	\$745 USD
FL3-FW-03S3C-C	SONY ICX414 1/2" CCD; color; 648 x 488; 76 fps; C-mount	\$845 USD	\$745 USD
FL3-FW-14S3M-C	SONY ICX267 1/2" CCD; mono; 1384 x 1032; 16 fps; C-mount	\$895 USD	\$845 USD
FL3-FW-14S3C-C	SONY ICX267 1/2" CCD; color; 1384 x 1032; 16 fps; C-mount	\$895 USD	\$845 USD
FL3-FW-20S4M-C	SONY ICX274 1/1.8" CCD; mono; 1624 x 1224; 15 fps; C-mount	\$995 USD	\$945 USD
FL3-FW-20S4C-C	SONY ICX274 1/1.8" CCD; color; 1624 x 1224; 15 fps; C-mount	\$995 USD	\$945 USD

Product Includes

- Tripod mount adapter
- FlyCapture image acquisition and camera control software
- Does **NOT** include lens or optics



Available Development Kits (recommended for first-time users)

DEVKIT-01-0001	\$100 USD	DEVKIT-01-0002	\$100 USD
▪ ACC-01-2006: 4.5m, 9-pin to 9-pin locking IEEE-1394b cable ▪ ACC-01-2007: 4.5m, 6-pin to 9-pin locking IEEE-1394a to 1394b cable ▪ ACC-01-1001: IEEE-1394b OHCI PCI Host Adapter 2 Port 800Mbps card ▪ ACC-01-3000: Hirose HR25 8-pin male GPIO connector with 1m wiring harness ▪ FlyCapture SDK (C/C++ API and device drivers) CD and Manuals ▪ Grasshopper Getting Started Manual		▪ ACC-01-2006: 4.5m, 9-pin to 9-pin locking IEEE-1394b cable ▪ ACC-01-2007: 4.5m, 6-pin to 9-pin locking IEEE-1394a to 1394b cable ▪ FWB-PCIE-01: FirePRO low profile single bus IEEE-1394b PCI Express card ▪ ACC-01-3000: Hirose HR25 8-pin male GPIO connector with 1m wiring harness ▪ FlyCapture SDK (C/C++ API and device drivers) CD and Manuals ▪ Grasshopper Getting Started Manual	

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Figure A.1: Quote for point grey firewire Flea3 cameras



products : [imaging](#) : [imaging lenses](#) : [fixed focal length lenses](#)

Megapixel Fixed Focal Length Lenses



- Lockable, Manual Focus and Aperture Control
- Working Distance from 100mm to Infinity
- Compact Size

Designed for high resolution, industrial C-Mount cameras, our Megapixel Fixed Focal Length Lenses are ideal for machine vision applications with demanding space constraints. These lenses complement the resolution of many megapixel cameras with 2/3" and smaller formats. Low distortion and high resolution enable these economically priced lenses to produce sharp, high contrast images compared to standard CCTV and compact lenses. Locking screws for the manual iris and focus allow these lenses to be used in high vibration environments. Typical applications include factory automation, inspection, and macro work. Lenses accept M30.5 filters. Maximum rear protrusion beyond the C-Mount is 1.1mm.

Description	Stock No.	Price	In Stock	Qty
5mm Megapixel Fixed Focal Length Lens	NT64-867	\$230.00	✓	BUY
8mm Megapixel Fixed Focal Length Lens	NT56-786	\$195.00	✓	BUY
Focal Length FL (mm)	8.00			Add to Wish List
Maximum Camera Sensor Format	2/3"			
Aperture (f/#)	F1.4-16C			
Field of View, 1/2" Sensor (°)	42.50			
Field of View @ Min Working Distance (mm)	87.00			
Working Distance (mm)	100 - ∞			
Maximum Diameter (mm)	33.50			
Length (mm)	28.20			
Maximum Rear Protrusion Beyond C-Mount (mm)	0.60			
Weight (g)	63.00			
Filter Thread	30.5 x 0.5mm			
RoHS	Compliant			
12mm Megapixel Fixed Focal Length Lens	NT56-787	\$195.00	✓	BUY
16mm Megapixel Fixed Focal Length Lens	NT56-788	\$195.00	✓	BUY
25mm Megapixel Fixed Focal Length Lens	NT56-789	\$195.00	✓	BUY
35mm Megapixel Fixed Focal Length Lens	NT64-868	\$195.00	✓	BUY
50mm Megapixel Fixed Focal Length Lens	NT56-790	\$195.00	✓	BUY
75mm Megapixel Fixed Focal Length Lens	NT64-869	\$255.00	✓	BUY

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Figure A.2: Quote for Edmund Optics Megapixel Fixed Focal Length Lenses

Preset Stage

<http://www.optosigma.com/products/manual-positioners/preset...>

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Preset Dovetail Stages / Preset Stage / 123-3540

Preset Stage, 65 x 65, XY Inch

Price	\$195.00
Quantity	0
Total	\$0.00

[Add to Request](#)

SpecsDocs and DrawingsRelated TypesQuestions

Dimensions a x b (mm): 65 x 65

Axes of Travel: XY

Travel (mm): 20 x 20

Dimensions tolerance: $\pm 0.2\text{mm}$

Finish: Black Chrome and Black Anodized

Sensitivity: 10 μ

Height , c (mm): 37

Load Capacity (kg): 7.5

Type: Inch

Material: Brass and Aluminum

Straightness: 30 μ

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Figure A.3: Quote for x-y stage for camera from Optosigam.



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4-6λ First Surface Mirrors



- Economically Priced
- Custom Sizes Available

These mirrors are coated with Enhanced Aluminum that provides >90% reflection in the visible. 95% reflection can be achieved depending on the wavelength of light used. Available in circles, squares and rectangles. Rectangular shapes are extremely useful in applications where the mirror must be mounted at 45° to produce a 90° bend in light path. Custom sizes are available for volume OEM requirements.

Can't find what you need? Get a quick [custom quote](#).

Description	Stock No.	Price	In Stock	Qty
4-6 Wave Mirror 81mm x 100mm	NT41-621	\$18.00		BUY
4-6 Wave Mirror 85mm x 101mm	NT32-362	\$21.00		BUY
4-6 Wave Mirror 102mm x 127mm	NT40-042	\$35.00		BUY
4-6 Wave Mirror 127mm x 178mm	NT41-320	\$45.50		BUY
4-6 Wave Mirror 127mm x 178mm	NT40-043	\$41.00		BUY
4-6 Wave Mirror 169mm x 194mm	NT41-405	\$41.00		BUY
4-6 Wave Mirror 200mm x 235mm	NT32-358	\$95.25		BUY
4-6 Wave Mirror 204mm x 254mm	NT41-321	\$84.50		BUY
4-6 Wave Mirror 204mm x 254mm	NT40-067	\$84.50		BUY
4-6 Wave Mirror 254mm x 254mm	NT46-653	\$84.50		BUY
4-6 Wave Mirror 254mm x 313mm	NT46-654	\$84.50	EMAIL	BUY
4-6 Wave Mirror 254mm x 313mm	NT32-248	\$84.50 		BUY
Dimensions (mm)	254 x 313			
Thickness (mm)	6.00			
Surface Accuracy (λ)	4 - 6 per Inch			
Substrate	Soda Lime Float Glass			
Coating	Enhanced Aluminum, R >95% from 450-650nm			
RoHS	Compliant			

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Figure A.4: Quote for Edmund Optics First surface mirror.

Bibliography

- [1] K. Nakamura, A. J. Gonsalves, C. Lin, A. Smith, D. Rodgers, R. Donahue, W. Byrne, and W. P. Leemans. Electron beam charge diagnostics for laser plasma accelerators. *Phys. Rev. ST Accel. Beams*, 14(6):062801, Jun 2011.
- [2] J. I. M. Botman, C. J. Timmermans, B. Xi, H. Heller, H. L. Hagedoorn, P. Brinkgreve, E. Dekkers, and J. Moerel. Dipole design for the euterpe storage ring. In *Proceedings of the 1993 Particle Accelerator Conference*, page 2892, Piscataway, NJ, 1993. IEEE.
- [3] Boling Xi, J.I.M. Botman, A.T.A.M. Derksen, A.H. Kemper, C.J. Timmermans, and H.L.Hagedoorn. Magnetic performance of the euterpe ring dipole. In *Proceedings of the 1994 Particle Accelerator Conference*, page 2232, Piscataway, NJ, 1994. IEEE.
- [4] Boling Xi. *Design Studies for the Electron Storage Ring EUTERPE*. PhD thesis, Eindhoven University of Technology, 1995.
- [5] Michiel de Moor. Characterization of euterpe dipole magnet as an energy spectrometer for laser wakefield acceleration. Technical Report CQT 2008-12, Eindhoven University of Technology, Eindhoven, Netherlands, December 2008.
- [6] Point Grey Research Inc. 12501 Riverside Way, Richmond, BC, Canada V6W 1K7, <http://www.ptgrey.com/>.
- [7] Edmund Optics. 101 East Gloucester Pike, Barrington, NJ 08007-1380 USA, <http://www.edmundoptics.com>.
- [8] Optosigma. 3210 S. Croddy Way, Santa Ana, CA 92704 USA, www.optosigma.com.
- [9] K. Nakamura, W. Wan, N. Ybarrolaza, D. Syversrud, J. Wallig, and W. P. Leemans. Broadband single-shot electron spectrometer for gev-class laser-plasma-based accelerators. *Rev. Sci. Instrum.*, 79(5):053301, 2008.
- [10] K. Makino and M. Berz. Cosy infinity version 8. *Nucl. Instrum. Methods Phys. Res. A*, 427:338, 1999.